Large Language Model Advisors for Smallholder Agriculture: From Technology Supply to Adoption Mechanisms

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Large language model (LLM) advisors are increasingly proposed for smallholder agriculture, yet the evidence supporting them is scattered across incompatible tasks, datasets, and evaluation conventions. Synthesising this literature is difficult because the field lacks shared benchmarks, shared metrics, and shared causal assumptions, so claims about usefulness, equity, and deployability cannot be compared across studies. We organise the review around five questions—accessibility constraints, adoption-effect identification, structural financial and infrastructural constraints, dataset and benchmark fragmentation, and metric skew—and analyse a structured evidence matrix of 84 papers together with a evidence base of 104 dataset nodes and 132 metric nodes, coding each paper's limitations, assumptions, and evaluation targets. The results are markedly asymmetric. Low-resource-language accessibility is reported and partly quantified, with measured word error rates and uneven ASR coverage across ten providers, whereas offline low-compute deployment is acknowledged as an open problem but never experimentally bounded. Only two of six adoption studies perform causal inference, both from one team on one cotton-sowing dataset using back-door adjustment, and the identification assumptions are never tested in smallholder conditions. Financial constraints bind across all farm classes, and fixed MRV and transaction costs exclude smallholders, contradicting the risk-neutral optimising-farmer assumption embedded in most adoption and contract models. No dataset is used as a benchmark by more than one paper, yielding zero de facto standards, and energy, latency, and adoption metrics each appear in only one or a few isolated studies while accuracy metrics dominate. We conclude that the field's central weakness is not missing capability but missing commensurability: without shared benchmarks, shared adoption metrics, and explicit testing of identification assumptions, LLM agricultural advice cannot be reliably compared, audited, or trusted.

Additional Key Words and Phrases: multi-agent collaboration, large language models, agent orchestration, evaluation benchmarks, agent security

1 INTRODUCTION

Introduction

Agricultural AI research has produced a dense and rapidly maturing body of methods for sensing, phenotyping, and prediction, spanning high-throughput stand counting and organ-level segmentation to 3D reconstruction and genomic prediction. These methods are typically validated on curated imagery or omics datasets collected under controlled or well-resourced conditions, and their reported accuracy has become the principal currency of progress in the field. In parallel, a distinct literature addresses the delivery of AI-based services to smallholder farmers through conversational advisors, voice pipelines, and multilingual assistants. The two literatures rarely meet: the former optimizes perception under benchmark conditions, the latter confronts connectivity, language coverage, and literacy constraints in the field. This separation is the central tension motivating the present review. Claims of "field readiness" for smallholder-oriented agricultural AI are routinely asserted, yet the evidence supporting them is distributed unevenly across technical, economic, and infrastructural dimensions, and the criteria by which readiness would be judged remain unspecified.

The asymmetry is visible in the accessibility literature. Language coverage for advisory systems has been documented and partly quantified: multilingual deployments report finite language sets and acknowledge that inclusive language support remains unresolved, while model-agnostic voice pipelines measure substantial word-error-rate reductions across providers and device conditions. By contrast, offline and low-compute inference for language-model advisors is acknowledged as an open problem rather than measured as an empirical boundary. Edge efficiency has been characterized only in narrow perception tasks, where hardware-aware compression demonstrates that deployment on microcontrollers is feasible for weed detection, but no comparable evidence exists for the advisory stack that smallholders would actually use. Readiness, in short, is reported where measurement is cheap and asserted where measurement is hard.

A second tension concerns the inferential basis for adoption claims. Only a small subset of studies implements formal causal identification, and those that do rely on observational backdoor adjustment with domain-knowledge defenses of unconfoundedness. The broader adoption literature rests on cross-sectional surveys, descriptive adoption rates, and predictive accuracy, which cannot separate technology effects from selection. Economic constraints are consistently identified as binding across farmer categories, with credit access among the strongest determinants of adoption, yet the profit-maximizing, risk-neutral agent assumed by contract and adoption models conflicts with the credit, tenure, and participation constraints that smallholders face. Evaluations of digital agriculture recommendations and digital tools for sustainable agriculture have begun to close this gap, but the identification assumptions themselves are rarely tested against smallholder realities.

A third tension is infrastructural and methodological at once. The dataset ecosystem is highly fragmented: in our own structured audit, no dataset is used as an evaluation benchmark by more than one paper, and the datasets that do exist are small, author-collected, and labeled at organ or function granularity rather than at the farm-level decision scale that advisory systems require. Evaluation practice mirrors this fragmentation. Precision metrics dominate, while energy, latency, and adoption metrics appear only sporadically and are defined incompatibly across studies. Foundational work on data's critical role in agricultural AI argues that data quality and representativeness gate downstream utility, and recent benchmarks for multimodal agronomic reasoning attempt to align evaluation with reasoning tasks rather than perception alone. Whether these efforts constitute a shared standard remains an open question.

This review departs from prior surveys in three respects. First, rather than cataloguing methods, we organize the literature around five question directions that jointly define field readiness: accessibility constraints of advisory systems, causal identification of adoption effects, financial and infrastructural constraints on use, dataset and benchmark maturity, and the alignment of evaluation metrics with deployment conditions. Second, we treat the distribution of evidence itself as a finding, distinguishing constraints that are reported and quantified from those that are merely acknowledged. Third, we connect the perception-oriented and service-oriented literatures, arguing that the true boundary of technical feasibility for smallholders is set not by peak accuracy but by the weakest measured link across language, compute, connectivity, data, and evaluation. Recent work on retrieval-grounded multilingual assistance, policy-grounded advisory architectures, and closed-loop co-pilots illustrates both the promise and the measurement gaps that this framing exposes.

Innovation

Existing reviews of agricultural AI tend to be organized by task or by model family, surveying advances in phenotyping, segmentation, and prediction with accuracy as the organizing axis. Reviews of digital agriculture for smallholders, conversely, tend to be organized by adoption theory or development outcomes, treating technology as a black box whose properties are summarized by adoption rates. Neither format supports the question this review poses, namely how far the technical supply side of agricultural AI is actually ready for smallholder field conditions, and where the constraints on that readiness lie. Our first innovation is to make readiness a structured, multi-dimensional object of analysis rather than an implicit summary judgment. We decompose it into accessibility, causal credibility, economic feasibility, data maturity, and evaluative fit, and we assess each dimension against the evidence rather than against stated intent.

The second innovation is methodological: we treat the presence or absence of measurement as primary evidence. By auditing which constraints are quantified and which are only named, we can distinguish a genuine feasibility boundary from a rhetorical one. This distinction surfaces findings that task-organized reviews cannot express, such as the contrast between extensively measured language accessibility and unmeasured offline inference for advisory systems, or between precision metrics that command community consensus and energy metrics used by a single study. The same logic applies to datasets: the absence of any benchmark shared across papers in our own structured audit is not a minor bibliographic detail but a structural obstacle to cumulative evaluation, and it directly undermines claims that a method is field-ready because it performs well on its own test set.

The third innovation is integrative. We connect perception-side deployment evidence, such as hardware-aware compression for low-power edge devices and lightweight 3D reconstruction pipelines, to service-side delivery evidence from advisory deployments. This connection reveals that the binding constraint on smallholder readiness is rarely the accuracy of any single component; it is the compounding of language coverage limits, connectivity and cost barriers, fragmented data, and evaluation metrics that do not measure what deployment requires. We further incorporate causal evidence on adoption and contractual design to show that technical feasibility and economic feasibility diverge systematically for smallholders, and that this divergence is itself measurable.

Contributions

- We define field readiness for smallholder-oriented agricultural AI as a five-dimensional construct and show that the dimensions are evidenced asymmetrically, with language accessibility quantified while offline low-compute inference for advisory systems remains unmeasured. - We provide a structured audit of the dataset and benchmark field, showing that no dataset is shared as an evaluation benchmark across papers and that existing datasets are small, author-collected, and labeled below the farm-decision granularity required by advisory services. - We document a systematic skew in evaluation metrics toward precision-oriented measures, with energy, latency, and adoption metrics used sporadically and defined incompatibly across studies. - We synthesize the causal-inference evidence on adoption and recommendation effects, identifying observational backdoor adjustment as the dominant design and showing that its identification assumptions are defended by domain knowledge rather than tested against smallholder conditions. - We establish that financial and infrastructural constraints are binding across farmer categories, with credit access among the strongest adoption determinants, and that these constraints conflict with the risk-neutral, unconstrained agent assumed in contract and adoption models. - We connect perception-side edge deployment evidence to service-side delivery evidence, arguing that smallholder feasibility is bounded by the weakest measured link across language, compute, connectivity, data, and evaluation rather than by peak model accuracy. - We identify concrete measurement gaps that future work must close, including offline inference benchmarks for advisory systems, shared evaluation datasets at farm-decision granularity, and adoption metrics with compatible definitions.

2 BACKGROUND AND PRELIMINARIES

This section fixes the vocabulary used throughout the review. We define five concepts that recur across the thematic chapters: agentic and multi-agent systems for agricultural knowledge work, neuro-symbolic grounding of perception and advice, the smallholder adoption decision as a modeled object, agricultural data governance and benchmarking, and cross-layer evaluation. For each concept we give a working definition, trace how the term is used in the literature under review, and state its relations to the other concepts. The definitions are deliberately operational: they are phrased so that later chapters can ask whether a given paper instantiates the concept, measures it, or merely assumes it.

This measurement gap reflects uncertainty about the object and scale of evaluation; clarifying these concepts provides a shared semantic basis for later discussion.

2.1 Agentic and Multi-Agent Systems in Agricultural Knowledge Work

An agentic system, as used here, is a software architecture in which one or more language-model-driven components plan, invoke tools, and act over multiple turns toward a task goal rather than emitting a single prediction. The term has entered agricultural computing through two routes. The first is scientific discovery and phenotyping pipelines, where an agentic framework coordinates data acquisition, model selection, and analysis steps that were previously chained by hand. The second is farmer-facing advisory, where a large language model is grounded in extension policy documents and constrained to produce answers consistent with local recommendations. Multi-agent denotes the case in which several such components, each with a narrower responsibility, are composed into a workflow; PHENOINTEL is an explicit instance, aligning agents to stages of a phenotyping lifecycle inside a web application [1]. Closed-loop co-pilots extend the idea from one-shot advice to iterative interaction in which the system observes the outcome of its own recommendation and revises it. We treat orchestration topology—the graph of which agent calls which, and where human or sensor input enters—as a design variable rather than a fixed property, because the reviewed systems differ substantially in whether control is centralized in a planner or distributed across peers.

Two qualifications matter for smallholder settings. First, agency presupposes an execution environment: tool calls, retrieval over a corpus, and persistent state. Where connectivity and compute are intermittent, the architecture must degrade to an offline or edge-resident mode, and the reviewed work reports this constraint unevenly. Second, language coverage is an architectural, not merely a linguistic, property: voice pipelines that are model-agnostic and language-agnostic are proposed precisely because the advisory layer must accept speech in low-resource languages before any agentic reasoning can occur [2]. We therefore treat the agentic layer as sitting above a perception and interface layer, which the next subsection defines.

2.2 Neuro-Symbolic Grounding of Perception and Advice

Neuro-symbolic refers to systems that combine learned representations with explicit symbolic structures—ontologies, rules, or evidence bases—so that outputs are both statistically derived and checkable against declared domain constraints. PHENONEST is the clearest statement of the concept in the reviewed corpus, pairing neural perception with an ontology-aware reasoning layer. The motivation is continuity between measurement and meaning: a segmented organ or an extracted trait is only useful to a downstream decision if it is typed against a shared vocabulary. Related work supplies the perceptual substrate. Multimodal root phenotyping couples 3D skeleton extraction with language analysis so that geometric structure and descriptive annotation are produced together [3]. Foundation models for 3D plant reconstruction reduce the cost of acquiring that substrate, moving cross-crop reconstruction from hours to seconds, while adaptive granularity selection addresses the resolution at which organs should be segmented across species. Synthetic data generation conditioned on the downstream task is a further route to closing the labeled-data gap that limits perception quality in under-resourced crops.

The relation between this concept and the previous one is directional: neuro-symbolic grounding supplies the typed observations that an agentic advisor reasons over, and it supplies the constraints that keep generated advice within agronomic bounds [4]. The relation to adoption is weaker and is the subject of a later chapter: grounding improves the plausibility of a recommendation but says nothing about whether a farmer will act on it. We flag this as a hand-off gap—the point at which a system stops producing validated advice and starts requiring a theory of the decision-maker.

2.3 The Smallholder Adoption Decision as a Modeled Object

We use adoption to mean the observable uptake of a technology or practice by a farm household, and adoption model to mean the formal object—regression, contract, or structural model—through which a study represents the choice to adopt. The empirical literature establishes that adoption responds to credit, extension contact, and risk exposure, with measurable welfare consequences. Survey evidence from digital technology uptake shows the same covariates operating on a different technology class. A distinct modeling tradition treats the interaction as a principal-agent problem: contracts for smallholder carbon farming are learned under moral hazard and adverse selection, making the incentive structure itself the object of design [5]. Forecast provision is a third entry point, where the question is whether improved information changes decisions rather than whether the forecast is accurate.

These strands share an assumption that deserves naming: the rational smallholder, a decision-maker who evaluates expected returns subject to a budget constraint. The assumption is analytically productive and empirically incomplete. Heterogeneity in land, tenure, liquidity, and infrastructure enters the reviewed studies largely as control variables or as sources of effect heterogeneity, not as structure that changes the form of the decision problem. Bounded rationality under binding constraints—satisficing, deferred adoption, and learning from neighbors—is rarely modeled explicitly, although the contract-theoretic work provides formal components that could carry it. The relation to the agentic and neuro-symbolic concepts is therefore asymmetric: technical systems produce recommendations under an implicit model of the farmer, while the adoption literature models the farmer without reference to the properties of the advising system.

2.4 Agricultural Data Governance and Benchmarking

Data governance, in this review, denotes the rules and infrastructure that determine who may access agricultural data, under what terms, and in what standardized form; benchmarking denotes the shared datasets, tasks, and metrics against which systems are compared. The reviewed corpus contains substantial dataset construction—multimodal root data [3], task-conditioned synthetic corpora, and reconstruction benchmarks enabled by 3D foundation models —but governance appears mostly as an implicit backdrop rather than an operational framework. Where policy enters, it enters as a grounding corpus for advice generation rather than as a data-sharing arrangement [4]. The consequence for smallholders is direct: data accessibility, language coverage, and the representativeness of benchmark crops jointly determine whether a system can be evaluated on the population it is meant to serve.

We distinguish governance from benchmarking because they fail differently. Fragmented benchmarks produce incomparable numbers; absent governance produces unavailable data and, eventually, unrepresentative benchmarks. Both are preconditions for the evaluation concept defined next, and both are treated in this review as cross-cutting rather than as properties of individual systems.

2.5 Cross-Layer Evaluation

Cross-layer evaluation is the practice of assessing a system along the full chain from perception through reasoning to farmer decision, using metrics that are defined at each layer and reported together. The reviewed work is concentrated at the lower layers: segmentation and reconstruction quality [6], prediction accuracy on agricultural tasks [7], and, less often, the operational cost of deployment. Energy, latency, and field robustness are reported sporadically, and adoption outcomes almost never appear in the same table as model accuracy [8, 4]. This distributional skew is itself a finding: when accuracy dominates the metric space, systems that are accurate but undeployable and systems that are deployable but modestly accurate become indistinguishable in the literature.

The concept ties the preceding four together. Agentic architectures [9, 1] and neuro-symbolic grounding [10] define what is being evaluated; adoption models [11, 5] define the endpoint that evaluation currently omits; governance and benchmarking [6, 2] define the conditions under which any comparison is meaningful. We adopt cross-layer evaluation as the organizing commitment of this review and return to its operationalization, including the metrics and reporting items it implies, in the evaluation chapter.

3 RELATED WORK

The literature on digital agriculture has grown along two largely separate tracks. One track asks whether a sensing, reconstruction, or prediction pipeline can be built and how accurately it performs; the other asks whether the resulting tools reach and benefit smallholders. Surveys of the first track are now abundant and increasingly specialized. Reviews of deep learning for image-based plant phenotyping [12] and of three-dimensional reconstruction from classical multi-view stereo through neural radiance fields and Gaussian splatting [13] codify a mature methodological canon, while bibliometric and systematic reviews of object detection with YOLO architectures and of remote-sensing data fusion from foundational to transformer-based techniques [14] map the applied computer-vision field. These surveys converge on a common conclusion: accuracy metrics such as mean average precision, intersection-over-union, and root-mean-square error are the field's shared currency, and architectural innovation is the dominant axis of progress. The second track is represented chiefly by systematic reviews of digital technology utilisation among smallholders [15], which catalogue adoption correlates but rarely connect them to the technical systems whose uptake they describe. The two tracks have not been integrated, and no survey we are aware of treats the translation from technical feasibility to field-level adoption as its organizing problem.

The comparability required by cross-layer evaluation must ultimately be grounded in whether specific research practices are organized around the same set of questions; the two strands formed in the digital agriculture literature—between technical performance and smallholder benefit—precisely provide an empirical field for testing this requirement.

A first major stream concerns three-dimensional sensing and organ-level reconstruction. Early field robots established that high-throughput, high-resolution three-dimensional phenotyping could be performed in situ, and subsequent work pushed reconstruction fidelity through neural radiance fields and through adaptive acquisition with radiometric calibration on unmanned ground vehicles. The introduction of organ-level point-cloud datasets under real field conditions [16] made quantitative comparison possible, and annotation-efficient segmentation methods together with boundary-aware panoptic segmentation addressed the labelling bottleneck. More recent contributions shift toward scalability and generalization: object-centric Gaussian splatting for fruit crops [17], foundation-model-based reconstruction that reduces capture time from hours to seconds [6], adaptive granularity selection for cross-species organ segmentation [18], and explicit diagnoses of the data–algorithm–computing gap. Our work shares this stream's commitment to rigorous three-dimensional measurement but treats reconstruction accuracy as an input to advisory and adoption questions rather than as an endpoint.

A closely related stream develops multimodal registration, tracking, and temporal modelling of plant growth. Multimodal image registration aligns disparate sensor modalities for phenotyping [19], leaf tracking recovers individual organ trajectories in top-down imaging, and multiview growth modelling captures developmental dynamics. Open platforms for two-dimensional temporal phenotyping [20] and interactive NeRF–SAM2 fusion frameworks lower the barrier to temporal analysis, while root phenotyping integrates three-dimensional skeleton extraction with language-based description [3]. Stomatal analysis systems demonstrate end-to-end automation of a single fine-grained trait. These works collectively establish that plant state can be measured repeatedly and reliably; they do not, however, examine how such measurements enter farmers' decisions. We build on their measurement guarantees while extending the pipeline toward decision support.

Generative and foundation-model approaches form a third, rapidly expanding stream. A scalable dif-

fusion foundation model for multi-crop phenotyping [21] and task-conditioned synthetic data generation [7] respond to the scarcity of labelled agricultural data, while synthetic and generated training data enable segmentation in weedy thermal imagery [22]. Large language models assist the procedural generation of editable, analysis-ready three-dimensional plant models [23], and super-resolution of satellite imagery guided by unmanned aircraft systems [24] fuses platforms of different resolution. Self-consistency prompting improves vision-language reliability for disease identification [25]. These contributions demonstrate that generative priors can substitute for expensive annotation; our work differs in using such priors to support reasoning and advice rather than reconstruction alone.

Genomic prediction constitutes a methodologically distinct stream with its own evaluation culture. Latent-factor approaches exploit high-dimensional secondary features [26] and hyperspectral time series, while kernel-based REML implementations [27] and density-power-divergence estimators improve robustness and heritability estimation. Interpretable deep models for crops [28], biology-informed neural networks [29], and transformer enhancements for genomic selection bring representation learning to breeding, complemented by genotype-to-phenotype prediction with nonlinear features and learnable group transforms for small structured datasets [30]. Reinforcement learning has been applied to breeding-program optimization [31], and prediction has been extended to non-crop species such as the honey bee and to rice improvement programs [32]. This stream is mature in its statistical machinery yet almost entirely decoupled from farm-level decision-making, a separation our framing seeks to bridge.

Field-level detection, counting, and segmentation constitute the most heavily benchmarked stream. Convolutional networks estimate plant stand counts at high throughput [33], and deep-learning corn stand counting has been integrated with precise robot control [34]. The Segment Anything foundation model maps smallholder field boundaries without training labels [35], while multi-growth-stage recognition addresses weed and crop discrimination across phenological stages. Uncertainty-guided attention and entropy-weighted losses improve seedling segmentation [36], and active learning has been critically assessed for its actual utility in image-based phenotyping. Transfer learning adapts models across species and conditions [37], and open pipelines such as MatchPlant operationalize single-plant detection from unmanned aerial vehicles [38]. We rely on this stream's demonstrated accuracy but note that its benchmarks are almost exclusively technical.

A parallel stream addresses deployment under hardware and energy constraints. Hardware-aware compression of YOLO detectors for low-power microcontrollers enables weed detection within tight energy budgets [39], and cost-effective cyber-physical prototypes bring monitoring within reach of resource-limited farms [40]. Networked-farm architectures propose tight integration of sensing and decision layers to address climate challenges [41], and digital twins support weather-driven decision-making under imperfect conditions [42]. These works make a compelling feasibility case for edge intelligence; they nonetheless evaluate success in latency, memory, and energy rather than in farmer outcomes, a gap our review foregrounds.

Conversational and multi-agent advisory systems represent the newest stream. PhenoAssistant couples conversation with multi-agent orchestration for automated phenotyping [43], and PHENOINTEL aligns a multi-agent web application with the plant-science lifecycle [1]. Agentic frameworks accelerate scientific discovery in phenotyping [9], and closed-loop LLM co-pilots support iterative analysis in digital agriculture [8]. Farmer.Chat scales AI-powered services for smallholders [44], retrieval-grounded multilingual assistance serves remote island communities [45], and policy-grounded architectures target agricultural extension in Ghana [4]. Fine-tuning and evaluation studies document persistent unsupported recommendations and generic advice [46], while the AIEP initiative reports technical learnings across five advisory deployments and explicitly acknowledges limited understanding of LLM behaviour in low-resource languages. This stream is closest to our concerns, yet it evaluates systems in isolation rather than as components of an adoption pathway.

A further stream examines economic and behavioural dimensions of advisory and market participation. Profit-aware advisory systems diagnose "economic blindness" in biologically optimized recommendations [47], and principal–agent models learn contracts for equitable smallholder carbon farming under moral hazard and adverse selection [5]. Studies of tractor service utilization identify financial constraints as the binding factor across farmer classes, and work on agricultural insurance access documents acceptability barriers among Ghanaian smallholders [48]. Digital extension research evaluates alternative delivery models for development outcomes, and analyses of adoption of ICT-in-agriculture innovations in Kenya trace determinants of uptake. These studies supply the behavioural vocabulary our synthesis adopts, but they seldom connect to the sensing and modelling pipelines that generate the advice being adopted.

Causal evaluation of digital recommendations is a small but important stream. Two studies apply observational causal inference with back-door identification to estimate average treatment effects of digital agriculture recommendations, cross-validating with linear regression, matching, inverse propensity score

weighting, and meta-learners [49]. Their identification assumptions—positivity, no unmeasured confounding, and correct functional form—are stated but not stress-tested against smallholder conditions, and the broader literature contains almost no mutually corroborating or conflicting causal estimates. We treat this stream as the methodological core of our argument, extending its concern with identification to the heterogeneity of smallholder constraints.

Data governance and AI-readiness form an emergent stream. Analyses of India's public agricultural data argue that adoption is limited less by model capacity than by data fragmentation and standardisation [50], and ethical and responsible AI recommendations for digital agriculture articulate fairness, transparency, and accountability obligations. Benchmark construction for agronomic reasoning from aerial imagery [51] and multimodal evaluation efforts respond to the absence of shared test beds. Our review observes that no dataset in the surveyed literature serves as a de facto standard across studies, which constrains cumulative comparison and motivates our emphasis on evaluation infrastructure.

Finally, a body of work situates technology within agroecological and food-security contexts. Reviews of ecological nutrient management [52] and agroecological technique adoption in Africa emphasize that practice change is constrained by resource endowments rather than information alone, and analyses of smallholder contributions to food security under climate stress frame the stakes of adoption. Yield prediction models tailored to smallholder decision support [53] and genomic prediction for rice breeding illustrate attempts to align modelling targets with farmer decisions. These contributions remind us that adoption is a practice-level phenomenon; our synthesis differs by treating them as boundary conditions on technical claims rather than as parallel literatures.

Across these streams, three blind spots recur. First, the translation mechanism between demonstrated technical feasibility and field-level adoption remains untheorized: accuracy-oriented surveys and adoption-oriented reviews [15] do not cite one another's constructs, and the causal studies that could bridge them are few and mutually isolated. Second, identification assumptions in adoption research are asserted rather than examined, and the tension between rational-agent models and the financial and infrastructural constraints documented empirically [5] is left unresolved. Third, evaluation is fragmented: accuracy metrics dominate, energy and latency are reported only within edge-deployment studies [39], and no shared benchmark supports cross-study comparison. Our review addresses these gaps by organizing the field around the translation from technical supply to adoption, making identification assumptions explicit, and proposing an evaluation frame that spans accuracy, deployability, and farmer-relevant outcomes.

- [13] A Survey on 3D Reconstruction Techniques in Plant Phenotyping: From Classical Methods to Neural Radiance Fields (NeRF), 3D Gaussian Splatting (3DGS), and Beyond
- [12] Computer Vision with Deep Learning for Plant Phenotyping in Agriculture: A Survey
- [15] Utilisation of Digital Technologies by Smallholder Farmers in South Africa: A Systematic Literature Review
- [49] Evaluating Digital Agriculture Recommendations with Causal Inference
- [54] Evaluating Digital Tools for Sustainable Agriculture using Causal Inference
- [55] Building AI-based advisory services for smallholder farmers: Technical learnings from the AIEP Initiative
- [1] PHENOINTEL: A Lifecycle-Aligned Multi-Agent Web Application
- [5] Learning Principal-Agent Contracts for Equitable Smallholder Carbon Farming under Moral Hazard and Adverse Selection
- [39] Hardware-Aware YOLO Compression for Low-Power Edge AI on STM32U5 for Weeds Detection in Digital Agriculture
- [51] From UAV Imagery to Agronomic Reasoning: A Multimodal LLM Benchmark for Plant Phenotyping

4 SURVEY METHODOLOGY

Of the 84 screened publications, 67 are cited in the synthesis; the remaining 17 informed screening and background and are not individually cited.

We assembled a corpus of 84 publications through multi-source retrieval, prioritizing arXiv and falling back to open academic indexes, with deduplication anchored on DOI and other persistent identifiers. Each record was then mapped to the five research questions and their 13 sub-questions, so that the scope of the review is fixed before any analysis begins.

Moving from a literature list to a reviewable evidence base still requires an explicit retrieval and classification procedure; the collection of 84 papers and their item-by-item mapping is precisely the first step in translating the aforementioned research questions into an operational analytical framework.

Refinement proceeded in two stages. First, inclusion required that a paper address at least one sub-question directly and report an evaluable method, dataset, or result; duplicates and non-archival variants were collapsed to a single canonical record. Second, the resulting evidence set was frozen prior to analysis, so that no paper could be added or removed once comparative work had started. This protocol separates retrieval from interpretation and keeps the evidential base auditable.

For evaluation, every paper was coded into six evidence fields: problem addressed, method, dataset and benchmark, metrics, limitations, and stated assumptions and constraints. These codes were used to build cross-paper correspondence links, allowing methods and results reported under different terminology to be compared on a common footing. The resulting evidence matrix supports three layers of analysis: construction of the matrix itself, per-sub-question comparison, and claim-level verification.

At the sub-question level we performed comparative, evolutionary, and trade-off analyses. Comparisons align papers that target the same sub-question; evolutionary analysis orders methods by the assumptions they relax or the constraints they add; trade-off analysis records where gains on one metric are reported alongside costs on another, such as accuracy against cost or generality against specificity. Each observation is tied to the specific papers that support it.

Conclusions were decomposed into checkable claims, and every claim is annotated with the publications that support it. Claims lacking direct support in the frozen evidence set were either qualified or excluded, and limitations and assumptions recorded in the six-field coding were carried forward into the synthesis rather than discarded. This keeps the distance between reported evidence and stated conclusion explicit and traceable throughout.

5 INSUFFICIENT FIELD READINESS OF AI TECHNOLOGY SUPPLY FOR SMALLHOLDERS

Once the eligibility screening and limitation recording of the evidence are complete, the next question to answer is: around which core issues do these screened and annotated publications actually revolve, and what kind of distribution pattern do their research orientations exhibit?

Having established the review methodology, this chapter turns to the technology supply side and examines whether the AI tools proposed for smallholder agriculture are actually ready for the conditions in which smallholders operate. The evidence reveals a consistent asymmetry: constraints such as low-resource language accessibility are documented with quantitative depth, whereas offline low-compute deployment, field robustness, and the transition from perception to farmer adoption remain acknowledged but largely unmeasured. We proceed from language accessibility, through edge-deployment evaluation and multi-agent/neuro-symbolic pipelines, to the boundary at which technical capability stops short of behavioral change—a boundary that motivates the conversion problem taken up in the next chapter.

5.1 Language Coverage as a Quantified Constraint, Offline Inference as an Unresolved One

To what extent have accessibility constraints under low-resource languages and offline, low-compute conditions been reported for conversational and LLM-based agricultural advisors? The evidence assembled here answers this question unevenly: language accessibility has been documented and partly quantified, whereas offline edge inference remains an acknowledged open problem with almost no direct measurement in advisory settings. This asymmetry, rather than a uniform gap, is the central finding, and it reframes field readiness as a property that varies sharply across constraint types.

Low-resource language accessibility is the most thoroughly evidenced dimension. The AIEP initiative states explicitly that understanding of LLM performance in low-resource languages and social-impact use cases remains limited, and that high-quality evaluation data are scarce. This is not a passing caveat: it identifies a missing evaluation substrate, since the absence of benchmarks prevents even the diagnosis of failure modes. The Gikuyu case illustrates the mechanism by which this scarcity propagates, as a well-performing fine-tuned MMS ASR degrades markedly once chained to machine translation, while the lack of a standard orthography complicates evaluation itself [55]. Language coverage is therefore constrained

not only by data volume but by the absence of shared written conventions against which systems could be scored.

Quantitative evidence sharpens this picture. A model-agnostic and language-agnostic voice pipeline evaluated ten ASR providers across Hindi, Telugu, and Odia and found coverage uneven, with no single model performing well on all three languages. The same pipeline reduced word error rates by 16–23% on cloud ASR, 5% on an on-device model, and 16–42% on multi-speaker audio. These figures establish that language accessibility is not merely discussed but measured, and that the gains are largest precisely where field conditions are hardest, namely multi-speaker recordings. In contrast, the most mature deployed system, Farmer.Chat, supports only six languages (Swahili, Amharic, Hausa, Hindi, Odia, Telugu, and English) and its authors concede that achieving language inclusivity remains a challenge. The gap between pipeline-level improvement and deployed coverage is thus a gap in breadth rather than in demonstrated technique.

A further disagreement concerns the processing route for multilingual advice. Farmer.Chat reports that generating directly in the source language underperforms a pivot strategy in which input is translated into English, processed, and translated back. AIEP, by contrast, observes that LLMs are increasingly able to merge translation and reasoning directly for languages such as Swahili and Hindi [55]. No controlled comparison adjudicates between these paths, so the choice currently rests on practitioner preference rather than evidence. This unresolved tension matters because pivot-based designs introduce a second translation stage whose errors compound, while direct generation depends on model coverage that the same literature describes as thin.

Offline and low-compute accessibility presents a markedly different profile. AIEP identifies hosting small speech models, or entire reasoning models, on user devices as the most thorough remedy for network latency, yet states that the performance and inference-time trade-offs of such models remain unclear [55]. The constraint is therefore named as an open problem rather than characterised. A retrieval-grounded advisor deployed for remote-island farmers makes the consequence concrete: its dialogue capability resides upstream, and if that service is unavailable the assistant cannot answer, with no local fallback model. Here the architecture converts a connectivity interruption into a total loss of function, which is a design-level rather than a model-level failure.

The only on-device measurement available concerns a different task. A YOLOv8n detector compressed through structured pruning and integer quantisation was deployed on an STM32 microcontroller, achieving 51.8 mJ per inference at 0.66 fps while shrinking the model from 9.95 MB to 850.97 KB [39]. These figures demonstrate that aggressive edge deployment is feasible for visual detection, but they cannot be transferred to conversational inference, whose memory and latency profile differs by orders of magnitude. The distinction is consequential: the existence of edge vision evidence may create an impression of readiness that the advisory literature does not support.

Infrastructure and affordability constraints are reported, but they are externalised rather than studied. A systematic review of digital technology use among South African smallholders lists unreliable electricity, high mobile data costs, poor connectivity, and language barriers as core constraints. These appear as contextual conditions rather than as variables manipulated in any evaluation. Farmer.Chat's user research similarly identifies data affordability as a principal problem for lead farmers and farmers in remote areas, and reports lower smartphone ownership among women and older users [44]. The equity dimension is thus documented at the level of user observation, without a corresponding methodological response.

Taken together, the coverage pattern is asymmetric in a way that carries direct implications for readiness claims. Language constraints are supported by error-rate reductions, provider comparisons, and explicit language inventories [44], whereas offline low-compute constraints rest on statements of uncertainty and on architectural limitations observed in deployment [55] [45]. The boundary of technical feasibility can therefore be drawn with reasonable confidence for language, but for offline operation it remains undefined rather than established. This difference in evidential status, not the magnitude of either gap, is what the field-readiness argument must accommodate.

Two qualifications limit the strength of these conclusions. First, quantified language evidence concentrates on a small set of languages, principally Hindi, Swahili, Gikuyu, Telugu, and Odia, leaving most low-resource languages of sub-Saharan Africa and Southeast Asia without direct measurement. Generality across the language dimension is therefore assumed rather than shown. Second, the partial coverage assigned to offline low-compute conditions rests on extrapolation from adjacent visual deployment, and no study reports accuracy, latency, and energy jointly for a small language model performing agricultural reasoning on farmer-grade hardware.

These findings indicate that the accessibility constraints most likely to determine real-world usability

are those least amenable to current evaluation practice. Language gaps can be expressed as error rates and coverage counts, and therefore enter benchmark culture readily; connectivity, power, and on-device inference resist this format because they depend on deployment context. The consequence is a systematic bias in what the literature can claim to know. Addressing it requires field trials under interrupted connectivity and constrained power, alongside joint accuracy–latency–energy profiling of compact models on realistic devices, rather than further refinement of already-quantified language metrics.

5.2 Dimensional imbalance among energy efficiency, latency, and field robustness in deployment-readiness evaluation

Do edge-deployment studies for smallholder agriculture actually report energy, latency, and field-robustness outcomes, and if not, what form does this evaluative gap take? The evidence assembled here indicates that these three dimensions are not uniformly absent but asymmetrically distributed: energy and latency appear as isolated, self-certifying engineering metrics in a small number of hardware-oriented studies, whereas field robustness is acknowledged as important yet rarely converted into comparable measurements. This asymmetry, rather than simple omission, is what allows technical feasibility to be overstated relative to genuine field readiness.

Energy consumption per inference is the sparsest of the three dimensions. Across the surveyed literature, only a single study reports it: a hardware-aware YOLO compression effort that reduces the model from 9.95 MB to 850.97 KB and parameters from roughly 3.2M to 960k, then measures 51.8 mJ per inference, 1510 ms latency, and 0.66 fps throughput on an STM32U575ZI microcontroller while retaining mAP50 of 0.517 and mAP50–95 of 0.403 [39]. Crucially, the same authors frame 0.66 fps as suitable for periodic patrol rather than continuous detection, and describe the accuracy as acceptable for many but not all field scenarios. Energy reporting therefore exists, but as an island tied to one compression pipeline rather than as a routine component of deployment evaluation.

Latency presents a more instructive case, because where it is reported the object of evaluation has shifted. The P-EVAL protocol formalises deployment assessment as a multi-objective problem spanning policy alignment, agronomic utility, P95 latency, and human-supervision load, and reports a reduction of P95 latency from 28.6 s to 12.9 s, a 55% improvement backed by a 59.6% cache reuse rate, alongside PAR of 0.94 and AUR of 0.95 [4]. This is the most rigorous treatment of latency in the corpus. However, its 1,240 evaluation cases are simulated queries, so the metric characterises service-side responsiveness under communication bottlenecks rather than end-to-end usability in a farmer's field. The indicator is present, but it points at a different question than field readiness.

A contrasting failure mode appears in conversational advisory systems, where latency is observed rather than measured. The remote-island multilingual assistant reports simple queries taking roughly 15–20 seconds and multi-tool queries about one minute, yet explicitly characterises these as anecdotal observations rather than measured results, and lists quantitative validation of answer quality, grounding fidelity, latency, and user satisfaction as future work [45]. The contrast with P-EVAL is sharp: both systems serve smallholders through natural-language interaction, but one embeds latency in a formal protocol while the other leaves it as informal impression. This divergence suggests that the presence of a latency number depends less on task difficulty than on whether the research design treats deployment cost as a first-class outcome.

Field robustness is the dimension where the gap is most systematic, and its causes are diagnosable rather than mysterious. PhenoNest validates only on 500 curated WisWheat samples for a single crop and concedes that absolute bounding-box token sequences introduce decoding delay that "may constrain real-time deployment on field-edge processing hardware" [10]. The admission is notable because it identifies a concrete architectural mechanism, sequence decoding, rather than a generic lack of data, as the obstacle to edge deployment. Robustness is thus not dismissed as unimportant; it is displaced by an engineering constraint the authors can name but not yet resolve.

Detection studies illustrate how accuracy metrics can mask fragility across operating conditions. MatchPlant achieves AP@0.5 of 85.9% for UAV-based single-plant maize detection, but mAP@0.5:0.95 falls to 40.4%, and validation covers only the early growing season from V1 to V6 [38]. The authors note that extending to mid- or late-season stages may require additional methodological development. The wide spread between the two IoU thresholds is itself diagnostic: a model that performs well under lenient localisation can degrade sharply under stricter criteria, and single-stage validation cannot reveal whether this degradation compounds over a season.

Imaging evidence makes the same point at the sensor level. A systematic study of multispectral camera exposure shows that automatic exposure yields lower R^2 and higher MAPE in the red-edge band,

and that highly reflective targets saturate at a reflectance of 0.50, producing what the authors describe as unrecoverable loss of detail; fixed exposure is preferred for spatio-temporal radiometric consistency. This finding is important because it locates a robustness failure upstream of the model, in the acquisition configuration itself. A pipeline validated under automatic exposure may report acceptable aggregate accuracy while silently discarding the spectral information on which downstream decisions depend.

Taken together, these cases reveal four distinguishable forms of evaluative absence. The first is metric substitution, where mAP, F1, or MAE stand in for deployment outcomes, as when the compression study foregrounds mAP50 of 0.517 as its principal result [39]. The second is mean-only reporting, where tail behaviour such as P95 or P99 latency is omitted; P-EVAL is a rare exception in reporting it at all [4]. The third is validation on simulated or curated samples, exemplified by the 1,240 simulated queries and the 500 curated wheat samples [10]. The fourth is explicit non-measurement, where authors state that deployment properties were not tested and defer them to future work [45].

A further pattern deserves separate mention: capability evaluation substituting for deployment evaluation. PhenoAssistant assesses tool selection and pipeline correctness in a conversational multi-agent phenotyping system, but does not report latency, energy, or end-to-end cost of the underlying model calls. Similarly, lifecycle-aligned multi-agent web applications for phenotyping organise functionality around workflow stages without profiling the resource envelope of deployment [1]. In both cases the evaluation is internally coherent yet answers a question about system capability rather than about whether the system can run acceptably under smallholder conditions.

The consequence of this distribution is asymmetric for smallholders specifically. Where hardware and sensor-based precision agriculture remains prohibitively expensive for most smallholders, and where rural broadband penetration is markedly lower than in urban areas, energy and latency cease to be optional optimisations and become hard gates on usability [15]. Yet it is precisely field robustness, the dimension that determines whether a system keeps working across seasons and environments, that is least often quantified. The mismatch is structural: the metrics most needed to judge whether a technology is usable are the ones least often reported, while the metrics most often reported describe performance on curated data rather than fitness for deployment.

Two boundaries on these claims should be stated plainly. First, the sparsity of energy and latency reporting may partly reflect extraction coverage rather than genuine research scarcity, so the characterisation of these metrics as isolated should be read as indicative rather than definitive. Second, the reported figures are not mutually comparable: 51.8 mJ per inference on a microcontroller and a 12.9 s P95 latency in a policy-grounded advisory architecture arise from different tasks, hardware, and workloads, and can serve only as task-specific reference points [39, 4]. What can be concluded is narrower but firmer: our evidence set contains no a shared protocol that would make energy, latency, and robustness jointly reportable and comparable across smallholder deployment contexts.

This absence of a shared protocol is what connects the evaluative gap to the broader question of field readiness. Because energy and latency are reported only where a study is explicitly engineering-oriented, and because robustness is displaced by named architectural constraints or deferred to future work, the published record cannot establish whether a given technology will function under the interrupted connectivity and constrained power that characterise smallholder settings. Closing this gap requires joint accuracy–latency–energy profiling on realistic devices together with multi-season robustness testing, rather than further refinement of metrics that were never designed to describe deployment.

5.3 From Perception to Recommendation: Where the Advisory Chain Stops Short of Adoption

Whether multi-agent and neuro-symbolic architectures actually carry smallholder advisory systems from perception through to a decision recommendation—and whether that recommendation ever reaches the point of adoption—frames the analysis that follows. The systems reviewed here demonstrate that the orchestration layer linking sensing to advice remains the least reliable part of the chain, and that even well-formed advice confronts adoption barriers that no architectural refinement can remove.

Multi-agent designs make the perception-to-recommendation path explicit, and in doing so they expose its fragility. PhenoAssistant decomposes phenotyping into specialised conversational agents and reports complete success on data-analysis tasks, yet its tool-selection success rate reaches only 70% against 98% for visual model selection [43]. The asymmetry is diagnostic: individual perception components perform near ceiling, whereas the routing decisions that assemble them into a coherent recommendation fail at roughly three times the rate. This suggests that the binding constraint in agentic advisory systems lies not in any single model but in the coordination logic that determines which capability to invoke and when.

Lifecycle-aligned architectures respond to this coordination problem by imposing verification structure

on the agent ensemble. PhenoIntel organises nine agents around a shared typed state and a sequence of validation checkpoints, reporting a 54% reduction in counting error and a detection mAP@50 of 0.96 while intercepting failures before they reach the generated report [1]. In contrast to the conversational routing of PhenoAssistant, this checkpoint discipline converts silent error propagation into explicit rejection, which is the more defensible behaviour when downstream users cannot independently audit the output. The comparison indicates that reliability in multi-agent advice depends less on agent count than on whether intermediate states are validated against typed expectations.

Neuro-symbolic coupling offers a second route to trustworthiness, though the evidence base for it is thin. PhenoNest is the only system in this set that genuinely instantiates the paradigm, decoupling a Qwen2-VL perception engine from an LLM-based logical engine and enforcing a binary consistency gate so that every asserted symbolic relation is backed by visual evidence [10]. The gate functions as a provenance mechanism: a relation either survives verification or is withheld. However, the framework is demonstrated on ontology-aware multimodal phenotyping rather than on farmer-facing advice, so its capacity to constrain recommendations under field ambiguity remains untested. A natural question is whether such gating scales to the open-ended queries that advisory settings generate.

Closed-loop systems push furthest toward decision execution, and precisely there the boundary with adoption becomes visible. A closed-loop LLM co-pilot reports a 35% shortening of production cycles, 18% energy reduction, and a further 67.9% saving from autonomous strategies [8]. Yet its outputs are hardware actuation commands rather than recommendations a farmer chooses to follow, and the authors themselves concede that human expert validation remains indispensable. The mechanism is therefore one of automated control within a managed facility, not of persuasion or enablement at the farm gate; the loop closes on machines, not on households.

Where systems do address farmers directly, the evidence shifts from architectural performance to interaction quality. Farmer.Chat and related multilingual, retrieval-grounded assistants extend conversational advisory to smallholder audiences [44, 45], while voice-pipeline work targets the language and channel barriers that exclude low-literacy users. Policy-grounded architectures such as Pezego-HITL add a human-in-the-loop layer intended to keep extension advice aligned with official guidance [4]. These systems improve the reach and intelligibility of recommendations, but their evaluations report engagement and satisfaction rather than verified changes in practice.

The AIEP initiative provides the clearest quantitative separation between advice quality and adoption. Across 800 farmers, advisory satisfaction reached an NPS of approximately 60, well above the 26 and 40 recorded for comparison regions, yet among those who did not adopt the advice, 50% cited unavailability of inputs or capital constraints and 33% found the information difficult to understand [55]. High satisfaction thus coexists with non-adoption, and the dominant reasons are material rather than informational. This pattern indicates that the perception-to-recommendation chain can function well while the recommendation-to-adoption link fails for reasons entirely outside the system's design space.

Structural constraint as the binding barrier recurs across adjacent agricultural services. In tractor service utilisation, 79.7% of farmers adopted while the remaining 20.3% cited capital limitation as the primary obstacle, and the adoption gap premium of GHS 471 versus GHS 312 per acre among non-adopters points to structural exclusion rather than unwillingness [56]. Agricultural insurance exhibits a sharper version of the same paradox: 90% of respondents considered the product useful but only 14% had access, with 64% lacking product knowledge and 23% facing regional unavailability, while merely 5% judged it too expensive. Read alongside the AIEP findings, these results suggest a general mechanism in which acceptability is high and accessibility is low—a configuration that advisory technology alone cannot resolve.

Causal evaluation frameworks constitute the only strand that quantifies the full advice-to-behaviour-to-output pathway. One framework estimates that sowing according to recommendations raises cotton yield by 12–17% [49], and companion work applies similar causal designs to digital tools for sustainable agriculture [54]. These studies establish that recommendations can move outcomes when followed, but they are observational, single-crop, and regionally bounded, and they function as assessment instruments rather than as systems that induce adoption. The gap they leave is precisely the one the agentic architectures above do not fill: none of the reviewed systems instruments the transition from a delivered recommendation to a realised practice.

Taken together, the evidence traces a chain that is technically articulated at its front end and institutionally severed at its rear. Multi-agent orchestration improves the assembly of perception into advice but remains error-prone at the routing step [43], verification checkpoints and symbolic gating contain that error without extending the chain's reach [1, 10], and the adoption literature shows that the terminal link is governed by input availability, finance, and comprehensibility rather than by recommendation quality

[55, 56]. The readiness gap for smallholders is therefore not solely a matter of model accuracy or agent coordination; it is a matter of systems that stop at advice and infrastructures that determine whether advice can be acted upon. This reframes the readiness question around the coupling between advisory delivery and the material conditions of the farm, a coupling examined next through the lens of deployment context and institutional support.

6 LACK OF CONVERSION MECHANISMS FROM TECHNICAL FEASIBILITY TO FIELD ADOPTION

Following the analysis of the placement of recommendations and infrastructure, the focus of the examination naturally shifts to the coupling relationship between recommendation delivery and farm material conditions, and from there proceeds to the analysis of deployment scenarios and institutional support.

Having established that the AI technology supply for smallholders remains insufficiently field-ready, the review now turns to the conversion mechanisms that would carry a technically feasible tool into sustained field adoption. The central tension is that technical feasibility and adoption-side evidence are produced on different objects: capability-oriented studies demonstrate that a tool works on agronomic or technical metrics, whereas adoption studies estimate whether adopters and non-adopters differ in measurable returns, and the two are almost never closed on the same system, population, and time window. This chapter therefore examines the causal-inference designs and identification assumptions behind adoption-effect claims, the systematic simplification of adoption costs and farmer decision rules, and the resulting object misalignment that breaks the evidence chain between supply-side feasibility and demand-side adoption. The chapter closes by tracing how this misalignment ultimately rests on an implicit rational-smallholder assumption that stands in unresolved tension with the heterogeneity of real smallholder populations.

6.1 Identification Assumptions and Their Limits in Smallholder Causal Inference

A central question for this chapter is which causal inference methods and identification assumptions underlie studies of adoption effects, and how thoroughly their validity conditions have been examined in smallholder settings. The evidence assembled here indicates that formal causal identification remains rare: among the studies reviewed, only two implement an explicit causal design, and both rely on observational backdoor adjustment whose assumptions are defended through domain knowledge rather than through the realities of smallholder farming. The remaining studies report descriptive or predictive associations and therefore cannot, on their own, support claims about the effect of adoption.

The two causally oriented studies share a common methodological architecture. Both adopt an observational causal inference framework with the backdoor criterion to identify the average treatment effect (ATE), and both cross-validate the estimate across five estimators—linear regression, matching, inverse propensity score weighting, and T- and X-learners [49, 54]. This multi-estimator strategy is a reasonable safeguard against specification dependence, since agreement across parametric and machine-learning estimators reduces the risk that a single functional form drives the result. However, estimator diversity addresses model uncertainty, not identification: all five estimators condition on the same observed covariate set, so they share whatever bias arises from unobserved confounding. The convergence of estimates therefore signals internal consistency rather than causal validity.

The empirical basis of these two studies is a single observational dataset on cotton sowing recommendations, with the second study analysing 171 plots comprising 51 treated and 120 control units [54]. Following the recommendation is reported to raise yield by 12–17%, corresponding to an ATE of 372–546 kg/ha against an observed mean of 3145 kg/ha, and the estimate survives four classes of refutation tests (placebo treatment, random common cause, random subset, and unobserved common cause) [49, 54]. These robustness checks are more informative than estimator agreement, because they probe the sensitivity of the estimate to plausible violations. Yet the unobserved-common-cause test only demonstrates that the estimate is stable under a simulated confounder of specified strength; it cannot establish that no such confounder exists, and the authors acknowledge that the no-unobserved-confounding assumption is in principle untestable [54].

The defence of the identification assumptions leans on mature agricultural environments rather than on smallholder conditions. The authors concede that it is impossible in practice to enumerate all interacting factors in the system and that they rely on established agronomic knowledge plus refutation tests to "minimise" bias [54]. This is a defensible position for a relatively homogeneous cropping system, but it shifts the burden of justification onto an assumption of contextual stability that smallholder settings rarely satisfy. Where plot sizes, intercropping patterns, labour availability, and input access vary widely within a village, the covariate set that suffices to block backdoor paths in one setting is unlikely to suffice in

another.

A more consequential tension appears in how the studies achieve identification at all. To make the backdoor criterion hold, one study treats cooperative management practices as constant and omits the agricultural knowledge (AK) node from the graph, while explicitly recognising that this restricts the external validity of the results [54]. The mechanism is straightforward: agricultural knowledge is plausibly both a determinant of adoption and a determinant of yield, so conditioning it away or holding it fixed removes a legitimate confounder at the cost of narrowing the population to which the estimate applies. Identification is thus purchased by sacrificing generalisability to the heterogeneous smallholder population that the advisory system is meant to serve. This is the clearest instance in the reviewed literature of a trade-off between internal and external validity being made explicit rather than concealed.

The companion study reaches a similar conclusion from the opposite direction. It attributes low external validity to the homogeneity of farmer management practices in the sample and states that the results cannot be generalised to farms following different conventions, leaving cross-season, cross-crop, and cross-site validation to future work [49]. Read together, the two studies describe a closed loop: the conditions that make identification credible—uniform management, a single crop, a single season—are precisely the conditions that make the estimate uninformative about the diversity of smallholder practice. The convergence is instructive because it arises from independent acknowledgements by the same research group, which strengthens rather than weakens the inference that the limitation is structural to the design.

In contrast, the studies that address smallholder populations directly do not attempt causal identification at all. A mixed-methods study of agricultural insurance in Ghana combines 100 questionnaires with qualitative interviews with 100 farmers, reports an availability and acceptability rate of only 14% against a perceived usefulness of 90%, and attributes 64% of non-adoption to insufficient knowledge and 23% to product unavailability [48]. These figures are striking, but they are descriptive and correlational: without an identification strategy, the study cannot rule out the endogeneity of knowledge and adoption, since farmers who seek information may differ systematically from those who do not. The 90% versus 14% gap is therefore best read as evidence of a demand-side puzzle rather than as an estimate of what would happen if availability or knowledge were exogenously increased.

A systematic review of climate-smart agriculture adoption among smallholders is even more explicit about its evidentiary limits. It states that it assessed frequency only and performed no bias assessment, and consequently its findings on adoption barriers carry no causal force. This candour is valuable, but it also means that the barrier catalogues commonly cited in the smallholder literature rest on counts of reported constraints rather than on contrasts between adopters and non-adopters. The gap between what such reviews can support and what policy readers often take from them is itself a mechanism in the failure to convert technical feasibility into field adoption.

Prediction studies occupy a third and frequently conflated category. A deep neural network model for maize yield prediction reports RMSE, MAE, and ARSE errors as low as 0. –0.02 t/ha [53]. Such precision describes the fit of a forecasting model, not the causal effect of a recommendation on yield, and the two quantities answer different questions: a model can predict accurately while the intervention it informs has no effect, and conversely. Treating low prediction error as evidence of adoption benefit is a category error that recurs across the advisory literature, and it inflates apparent support for interventions whose causal status is not represented in our evidence set tested.

The broader corpus reinforces this picture of an evidence base organised around feasibility and uptake rather than effect. Reviews of digital technology utilisation among South African smallholders and of adoption in Indonesia document patterns of use and barriers to uptake without isolating causal effects [15, 57], while work on building AI-based advisory services for smallholders reports technical learnings from deployment rather than identified impacts [55]. Similarly, analyses of the data preconditions for AI in agriculture and of profit-aware advisory design address what is needed to deliver recommendations, not whether delivered recommendations change outcomes. The cumulative impression is of a field that has invested heavily in the supply side of advice and comparatively little in credible evaluation of its demand-side effects.

Two further strands illustrate how identification concerns surface in adjacent problems. Work on principal–agent contracts for smallholder carbon farming explicitly models moral hazard and adverse selection [5], showing that incentive design under unobserved farmer effort is tractable when the information structure is stated formally; this stands in contrast to adoption studies that leave confounding implicit. At the same time, methodological work on self-consistency in vision-language models and on biology-informed neural networks for genomic prediction advances measurement and representation rather than causal effect estimation, as does scalable organ-level plant segmentation [58]. These contributions improve the inputs to advisory systems but do not close the evidentiary gap between recommendation and realised benefit.

Taken together, the reviewed evidence establishes a clear pattern: causal identification of adoption effects has been attempted only in a narrow, homogeneous cotton setting, where it depends on assumptions—no unobserved confounding, constant management practice, omitted knowledge—that are defended by agronomic domain knowledge and that the authors themselves recognise as limiting external validity [49, 54]. Studies that do reach heterogeneous smallholder populations remain descriptive, and prediction studies measure fit rather than effect [48, 59]. The conditions under which identification holds in smallholder settings have therefore been discussed only to the extent of acknowledging that they may not hold, not to the extent of testing where and for whom they fail. This unresolved boundary motivates the next section, which turns from whether adoption effects can be identified to why, even when they are positive, they so rarely translate into sustained field-level conversion.

6.2 Adoption Cost Representation and the Systematic Simplification of Farmer Decision Models

How do the reviewed studies represent adoption costs and the economic decision logic of farmers, and do these representations conceal systematic simplifications? Across the corpus, adoption is overwhelmingly encoded as a binary treatment or an assumed consequence of technical superiority, while the costs, risks, and contractual frictions that condition smallholder behaviour are either omitted or relegated to background narrative. This pattern is not incidental: it recurs across causal-inference evaluations, diagnostic reviews, and optimisation-based advisory designs, and it constitutes one of the principal reasons why technically feasible interventions fail to convert into sustained field practice.

The most direct simplification appears in causal evaluations of digital recommendations, where adoption is treated as a binary treatment and farmers are implicitly assumed to sow whatever the model recommends. The evaluation framework of [49] establishes that following sowing recommendations raises cotton yields by 12–17%, yet the estimand captures the effect of recommendation uptake rather than the decision to uptake. Consequently, the analysis cannot distinguish a farmer who declined because the input bundle was unaffordable from one who declined because the recommendation was never received. The causal quantity is well identified, but the behavioural mechanism that governs whether the treatment is ever assigned in practice remains outside the model, which limits what such estimates can say about conversion at scale.

Survey-based evidence indicates that the omitted costs are neither small nor uniformly distributed. A systematic review of smallholder digital technology use in South Africa groups constraints into end-user, service-provider, and technology-characteristic categories, and reports ICT costs described as prohibitive, expensive mobile data, absent infrastructure, and language barriers as recurrent themes [15]. The same review observes that farmers cluster around tools of low technical complexity, which suggests that the cost barrier operates jointly with usability rather than as an isolated price effect. This co-occurrence matters for interpretation: a binary adoption variable absorbs both the affordability constraint and the complexity constraint into a single undifferentiated contrast.

A complementary diagnosis locates the binding constraint upstream of the farmer altogether. [50] argues that the principal bottleneck to scaling agricultural AI in India is data infrastructure rather than algorithmic capability, and estimates that 86% of smallholders lack the capacity to compensate for weak data infrastructure, exposing them to systematic exclusion from data-driven tools. Read against the survey evidence, this points to a two-layer exclusion: farmers face direct cost barriers at the point of use, and indirect exclusion arising from the data conditions under which tools are built. Simplifications at the modelling stage therefore compound, rather than merely reflect, asymmetries that already exist in the field.

The sharpest contrast in the corpus comes from work that does explicitly model adoption cost. A principal–agent simulation incorporating fixed measurement, reporting, and verification (MRV) costs plus area-dependent variable costs, together with private farmer costs and unobservable effort, finds that a profit-maximising pooled contract yields an 87.7% adoption rate among large farms but only 8.2% among smallholders [5]. The magnitude is instructive: the exclusion gap exceeds what cost structure alone would predict, and the authors attribute the residual to the interaction between profit maximisation and the pooling constraint. Where adoption cost is modelled, smallholder exclusion emerges as a structural property of the contract rather than a residual to be averaged away.

Even this more explicit treatment carries acknowledged simplifications. The farmer model in [5] assumes risk neutrality, abstracts from credit, tenure, and participation costs beyond MRV, and specifies myopic best responses within a standard Stackelberg setting. The authors concede that the omitted frictions bind smallholders more tightly than large farms, and therefore present their exclusion estimate as a conservative lower bound. This is a rare instance of a study stating the direction of its own bias; most of the corpus leaves the sign of the simplification unexamined, which makes cross-study comparison of adoption magnitudes

hazardous.

A distinct but related simplification concerns what the objective function rewards. [47] characterises conventional crop recommendation as economically blind, optimising biological yield while ignoring market price and thereby steering farmers toward agronomically sound but financially infeasible decisions; incorporating price into a profit-aware model raises random forest accuracy to 99.3%. The contrast with yield-only evaluation is direct: the same recommendation can be correct under an agronomic criterion and wrong under a financial one. Profit awareness thus narrows one gap, yet the model still stops at the recommendation boundary and does not represent the farmer's cost of acting on it.

Cost appears in a still weaker role in diagnostic and prediction-oriented work. A vision-language pipeline for precision agriculture reports diagnostic accuracy rising from 82.2% to 87.8%, symptom accuracy from 38.9% to 52.2%, and treatment accuracy from 27.8% to 43.3%, while cost enters only as an incidental benefit in the form of 15–30% chemical reduction from targeted spraying [25]. Similarly, a biology-informed neural network motivated by reducing expensive large-scale field trials reports rank-correlation gains up to 56% and prediction-error reductions of 75%, with cost confined to background narrative and deployment driven entirely by genotype inputs [29]. In both cases the cost that is discussed is the cost of producing the prediction, not the cost to the farmer of using it.

Taken together, the corpus exhibits a consistent asymmetry: production-side costs are quantified with care, whereas adoption-side costs are either assumed away or acknowledged only qualitatively. The binary-treatment design of [49] and the constraint inventory of [15] describe the same phenomenon from opposite ends—one measures effects conditional on uptake, the other catalogues why uptake may not occur—and neither closes the loop. The principal–agent formulation of [5] is the only design in this set that endogenises the adoption decision, and its result suggests that closing the loop changes the conclusion substantially rather than marginally.

The direction of the resulting bias is therefore not neutral. If omitted frictions bind smallholders more tightly, as [5] argues, then evaluations that abstract from them will overstate achievable adoption among the population that matters most for conversion. The data-infrastructure exclusion identified by [50] reinforces this reading, since it operates before any farmer-level decision is made. A natural inference is that reported adoption effects should be read as upper bounds on field-level conversion for smallholders, although the corpus rarely states this qualification explicitly.

The boundary conditions of these simplifications remain largely untested. Risk neutrality, the absence of credit and tenure constraints, and myopic response are plausible approximations for large commercial operations and considerably less plausible for resource-constrained smallholders, yet no study in this set varies these assumptions systematically to locate where the approximation breaks. Profit-aware recommendation [47] improves the objective but does not model the farmer's capacity to finance the recommended input bundle. Whether the simplification is benign or decisive thus remains an empirical question rather than a settled modelling convention.

In sum, adoption cost and farmer decision logic are represented in three descending levels of fidelity: endogenised within a contractual model, reduced to a binary treatment, or displaced entirely by production-side cost narratives. Only the first level produces estimates in which smallholder exclusion is a modelled outcome rather than an unexplained residual, and even that level omits frictions whose omission is acknowledged to bias estimates downward. This gradient in representational fidelity, rather than any single missing variable, is what allows technically positive results to coexist with persistently low field adoption. It also sets up the question of whether the conversion gap is fundamentally a matter of missing decision structure or of institutional arrangements that no farmer-level model can supply.

6.3 Feasibility–Adoption Cross-Validation and the Missing Behavioural Bridge

Does the technical feasibility established on the supply side actually corroborate the causal adoption evidence gathered on the demand side, and if not, where precisely does the bridge break? A direct cross-validation between the two bodies of evidence is largely absent: the supply-side systems and the demand-side causal studies operate on different technical objects, so that no single study in the current literature connects the feasibility result of one system to the adoption outcome of that same system. The biology-informed neural network (BINN) reported by [29] achieves up to a 56% improvement in test-set rank correlation and a 75% reduction in prediction error on maize TWAS and synthetic metabolic benchmarks, yet every validation remains at the modelling level and never reaches field adoption or economic return. Likewise, the plant segmentation platform of [58] delivers reproducible benchmarks (SPVConv at 86.66% semantic mIoU, SPVFormer at 64.48% instance AP) but is evaluated on five academic point-cloud datasets with no industrial-scale or adoption evidence. These are infrastructural tools whose demonstrated

feasibility is real but whose downstream uptake is simply not measured.

The one genuine bridge in the literature comes from a pair of sister papers on the same cotton sowing advisory system, which combine causal identification with deployment among real collaborating farmers. [49] report that following the recommendations raised yield by 12–17% (372–546 kg/ha against a mean of 3145 kg/ha), and the result survives four refutation checks—placebo, randomised control, random subset, and unobserved confounder—as documented across the companion study [54]. This is the only case in which a technical feasibility claim and an adoption-side causal claim are anchored to the same system, and it therefore functions as an existence proof that the bridge can be built rather than as evidence that it commonly is. The contrast with the BINN and segmentation work is instructive: those studies optimise representational fidelity, whereas the cotton case optimises a decision that a farmer can act on, and only the latter admits a causal adoption estimate.

A closer reading of that bridge, however, reveals that it is narrower than its headline number suggests. [49] treat the chain from evidence to trust to adoption as a motivational assumption rather than an object of test: the paper estimates only the average treatment effect of following advice on yield and does not measure trust, willingness to adopt, or diffusion. The authors themselves flag the constancy of agricultural practice—knowledge acting only through practice—as a source of limited external validity. The causal estimate therefore establishes that advice can move yield when followed, not that the evidence-to-trust-to-adoption mechanism operates as theorised. This is the first and most consequential break point: the demand-side causal result is conditional on compliance, and compliance itself is left unmodelled.

The second break point lies in the determinants of non-adoption, which the demand-side evidence attributes largely to structural and supply-side constraints rather than to willingness alone. In the tractor-service study of [56], adoption exhibits a gap premium of GHS 159 per acre (ATT = GHS 471 versus ATU = GHS 312), and farm size shows a counter-intuitive negative effect that the authors trace to fleet-capacity and other supply-side limits. This finding complicates any reading in which adoption follows straightforwardly from perceived benefit: farmers who would gain most are precisely those constrained by service availability, so the binding obstacle sits upstream of the individual decision. The implication is that a technically feasible advisory cannot by itself close the gap when the complementary service layer is capacity-constrained.

A third break point concerns how supply-side quality uncertainty erodes the trust on which adoption depends. [60] show that an abundance of poor-quality, hard-to-verify weather forecasts produces a "race to the bottom" and depresses demand for future improved products. The mechanism is a reputational externality: when farmers cannot distinguish good forecasts from bad, quality signals collapse and the market rewards the cheapest offering. This connects the supply and demand sides in a way the feasibility studies do not anticipate—technical quality is necessary but not sufficient, because unverifiable quality actively destroys the trust that the evidence-to-adoption chain requires. The cotton advisory's refutation checks guard against confounding in the yield estimate but say nothing about this reputational channel.

Broader adoption reviews reinforce the picture of a multi-dimensional socio-technical determination that no single feasibility result can satisfy. Syntheses of smallholder adoption consistently identify digital literacy, cost, infrastructure, and institutional support as joint determinants, with perceived usefulness itself a function of how observable the benefits are [57]. Read against the cotton case, this suggests that the 12–17% yield gain is observable and hence adoptable precisely because it is a practice-level outcome, whereas the gains from a better genomic predictor or a finer segmentation mask are not observable to the farmer and therefore lack an adoption pathway. The distinction is not one of technical merit but of whether the benefit is legible at the point of decision.

Taken together, the supply-side and demand-side conclusions do not cross-validate; they run on parallel tracks that meet at exactly one system. The bridge breaks at three coupled points: compliance is assumed rather than modelled [49], structural service constraints bind before individual choice [56], and unverifiable quality undermines the trust that mediates between evidence and use [60]. Each break point is behavioural or institutional rather than technical, which indicates that the conversion gap is not a deficiency of model accuracy but a missing decision structure connecting a validated prediction to an actionable, trustable, and serviceable choice. This reframes the problem for what follows: the question is less whether the technology works than whether the arrangements exist that let a working technology become a decision a farmer can actually make.

7 TENSION BETWEEN SMALLHOLDER HETEROGENEITY AND THE RATIONAL SMALLHOLDER ASSUMPTION

The fact that a technology can operate does not automatically mean that farmers can make decisions on that basis; for validated predictions to be transformed into adoptable choices, it is also necessary to examine the organizational conditions and institutional arrangements that support this transformation.

Having established that technically feasible solutions fail to reach fields, the review now turns to the analytical lens through which adoption is theorized. The rational smallholder assumption treats farmers as homogeneous expected-utility maximizers, yet the evidence reveals structural constraints—financial, infrastructural, and informational—that bind across all farm categories and systematically reshape what optimization can mean. This chapter therefore examines how smallholder heterogeneity and these binding constraints jointly challenge the rational-actor premise, and it develops the case for modeling bounded rationality under constraint as an explicit, operational alternative. That argument exposes a further dependency: such modeling requires shared data definitions and governance mature enough to represent heterogeneous agents and their constraints, a condition the next chapter assesses.

7.1 Financial and Infrastructural Constraints as Unmodelled Frictions in Adoption Theory

Which limitations in the empirical literature characterise financial and infrastructural constraints, and how do these constraints collide with the assumptions embedded in models of the rational smallholder? The evidence reviewed here indicates that financial and infrastructural frictions are not peripheral modifiers of adoption behaviour but first-order binding constraints that the canonical profit-maximising, credit-unconstrained, infrastructure-neutral adoption model cannot accommodate. This section traces that collision across four fronts: the empirical primacy of credit constraints, the residual heterogeneity that persists once financial barriers are relaxed, the acceptability–accessibility gap in risk-transfer instruments, and the infrastructural and data deficits that condition digital tool use.

Financial constraints emerge as the most universally binding limitation across smallholder categories, and their empirical ranking directly contradicts the assumption of a credit-unconstrained, risk-neutral optimiser. In a Ghanaian maize system, Kendall's coefficient of concordance places finance at mean rank 1.73, the top-ranked constraint across all farmer groupings rather than a burden concentrated among the poorest [56]. The same study's profitability analysis makes the conflict explicit: the adoption model it tests presumes no credit or tenure costs, and the authors list this presumption as a formal limitation rather than a modelling convenience [56]. The tension is therefore structural, not incidental—the model's behavioural core assumes away precisely the constraint that farmers rank first. A parallel Ethiopian analysis strengthens the point with adoption-side evidence: roughly 50% of adopters had access to credit against only 23% of non-adopters, and credit availability is significantly positive for both adoption probability and per-capita consumption welfare [11]. Read together, these findings suggest that credit access operates as a gatekeeper variable whose omission from adoption models biases both predicted adoption and predicted welfare gains.

However, relaxing the financial constraint does not restore the homogeneity that the rational smallholder assumption implies, and this is where the most instructive conflict surfaces. A principal–agent analysis of carbon farming contracts reports that a profit-maximising aggregator achieves 87.7% of attainable adoption on large farms but only 8.2% on small farms under a residual adoption share metric; crucially, the residual gap exceeds what fixed-cost structure alone can explain and is attributed to the interaction between profit maximisation and a single pooled contract [5]. The mechanism is contractual rather than purely financial: a pooled instrument calibrated to the average participant systematically under-serves the heterogeneous smallholder tail. This interpretation is corroborated by heterogeneity-aware impact estimates showing that non-adopters, were they to adopt, would realise an average treatment effect on the untreated of GHS 312 per acre—below the GHS 471 per acre observed for current adopters—because they lack complementary inputs, experience, and scale [56]. Financial constraint and complementary-asset constraint are thus coupled: removing the credit barrier alone leaves the complementary-input deficit intact, which is precisely the interaction the credit-neutral model cannot represent.

A second, subtler conflict appears in risk-transfer instruments, where stated acceptability and realised access diverge sharply. Among Ghanaian smallholders, approximately 90% regard agricultural insurance as useful, yet only 14% have ever purchased it; the dominant reported barriers are insufficient knowledge (64%) and product unavailability (23%), with cost cited by merely 5% [48]. This acceptability–accessibility gap inverts the standard demand-side story: the binding friction is supply and information, not willingness to pay. For adoption theory, the implication is that a model treating insurance as an available, priced option mislocates the constraint; the smallholder is not declining an offered contract but is frequently not offered

one at all. The gap also illustrates why single-construct behavioural theories underperform in this setting, since the missing variable is institutional availability rather than individual preference.

Infrastructural and digital-access constraints extend the same logic to technology-mediated inputs, and here the literature is unusually consistent about the direction of causality. A systematic review of South African smallholders identifies usage costs—mobile data fees, poor signal, absent electricity, and language barriers—as systemic constraints that operate alongside land availability and diseconomies of scale to impede digital tool use [15]. In the Indian context, the principal bottleneck for smallholder AI adoption is characterised as a data-infrastructure deficit rather than algorithmic capability: smallholders constitute 86% of farm households yet lack the capacity to compensate for weak data infrastructure [50]. The contrast with algorithm-centric narratives is instructive—investment in model sophistication yields diminishing returns where the binding constraint lies in connectivity, electricity, and data generation. These findings indicate that infrastructure enters adoption as a precondition rather than a cost term, which is incompatible with models that treat technology as uniformly accessible at a given price.

The theoretical literature itself registers this incompatibility, and its self-diagnosis is a useful anchor for the argument. A review of Indonesian smallholder digital adoption concludes that dominant frameworks—DOI, TAM, and UTAUT—explain adoption only partially: TAM is characterised as excessively individualist, while DOI neglects economic risk and structural barriers, so that adoption cannot be modelled as purely individual rational deliberation [57]. This is a direct challenge to the rational smallholder premise from within the adoption-theory canon, and it aligns with the empirical pattern in which structural constraints outrank attitudinal ones. The convergence is notable because it arises from two independent directions: econometric constraint rankings and conceptual critique of behavioural frameworks.

Taken together, the evidence supports a qualified conclusion: financial and infrastructural limitations are best understood not as parameters to be estimated within the rational smallholder model but as boundary conditions that determine whether the model applies at all. The credit-primacy result [56], the welfare relevance of credit access [11], the contractual residual on small farms [5], the insurance acceptability gap [48], and the infrastructural and data deficits [15, 50] converge on a single diagnosis: the constraints are coupled, and relaxing one without the others yields sub-additive gains. This coupling is the mechanism behind the persistent heterogeneity that motivates the next section's examination of how complementary assets and institutional access jointly stratify smallholder response to identical technologies.

7.2 Bounded Rationality under Structural Constraints: Theoretical Constructs for Adoption Modelling

Does the literature explicitly model bounded rationality under constraints, and which theoretical constructs can be borrowed for that purpose? The evidence assembled in this review supports an affirmative answer, though the maturity of that modelling is partial and fragmented: the field has moved from treating constraints as exogenous covariates in an expected-utility calculus toward treating them as structural features that redefine the farmer's optimisation problem itself, and a usable set of theoretical constructs is already visible across a small number of formal models and behavioural frameworks.

The older paradigm remains clearly legible in the adoption econometrics that still dominates the empirical literature. In an Ethiopian study of improved wheat, adoption is defined by the expected-utility rule that a farmer adopts when $U_a > U_{na}$, and a Double-Hurdle specification then enters credit access (2.002^{**}), extension contact (1.201^{**}) and plot distance (-0.254^{***}) as exogenous explanatory variables [11]. A Ghanaian tractor-service study rests on a triple foundation of profit maximisation, utility maximisation and Rogers-style diffusion, estimating determinants and profitability premia through MVP and PSM [56]. In both designs heterogeneity is rendered as a distribution of observable covariates, and a constraint is a coefficient rather than a force that alters the decision problem; this is precisely the starting point any shift must cross.

Against this background, the most fully computable realisation of constrained bounded rationality is a POMDP formulation of carbon-farming contracts, which explicitly models private adoption costs drawn from a uniform distribution and unobservable to the aggregator (adverse selection), unobservable effort (moral hazard), and a partially observed farmer population of which the aggregator sees only aggregate statistics [5]. The model further assumes that farmers best-respond myopically each season with a short planning horizon, drawing on behavioural evidence of smallholder short-sightedness. This is a qualitative rather than quantitative departure: the information structure, the incentive structure and the temporal horizon are all rewritten, so that the farmer is no longer a fully informed multi-season optimiser but a locally optimising agent inside an incompletely observed system.

The consequence of internalising constraints is not incremental. Within the pooled-contract setting, a profit-maximising aggregator does not merely inherit but amplifies exclusion: large farms realise 87.7%

of attainable adoption while smallholders realise 8.2% [5]. Because that residual exceeds what the cost structure alone can explain, and because a counterfactual in which monitoring costs are allocated purely by area makes the size gradient disappear entirely, the result indicates that treating constraints as exogenous covariates systematically understates exclusion. An analogous endogenous ordering appears among Ghanaian maize farmers, where financial constraints rank first across all size and gender classes, yet smallholders rank insufficient government support second while larger farmers rank it fourth [56]. Constraint hierarchies, in other words, vary systematically with group position rather than behaving as exogenous noise.

The most direct empirical signature of bounded rationality is a cognition–adoption paradox. Among Ghanaian smallholders, roughly 90% regard agricultural insurance as useful but only 14% actually obtain it, and the leading barrier is insufficient knowledge (64%) rather than price, with only 5% citing cost as prohibitive [48]. A pure price or utility-maximisation account cannot accommodate a gap of this shape, since the perceived value is high and the stated price objection is marginal; the pattern is better explained by information incompleteness and cognitive constraints operating on the conversion from belief to action.

Risk and loss preferences supply a second construct, and one with direct empirical backing. Farmers characterised by risk and loss aversion adopt significantly later, a pattern documented for Chinese cotton growers facing Bt cotton and summarised in the behavioural adoption literature [41], while risk preference also shapes the dynamics of insurance uptake [48]. The mechanism is a reconstruction of the payoff function rather than a shift in its level: under limited financial buffers, a technology with a favourable mean return may still be rationally deferred because its downside threatens consumption. A related reconstruction appears in digital extension research, which argues that farmers decide according to return on investment rather than yield maximisation, with ROI itself modulated by risk and preference, a consideration that conventional extension advice almost never incorporates.

Social learning, trust and learning costs form a third cluster of constructs. Adoption requires learning, learning is costly, and networked diffusion is cheaper than traditional extension, while conditional cooperators contribute only when they trust that others will do likewise [41]. The Indonesian ASTAD framework operationalises this layer through economic risk-buffering capacity, digital learning elasticity, a collective trust infrastructure and a peer demonstration effect, and it explicitly rejects the premise that adoption is the product of individual rational deliberation [57]. A systematic review of South African smallholders reaches a compatible conclusion, locating digital uptake within socio-technical rather than purely individual determinants [15].

These constructs, however, are distributed across literatures that have not been integrated into a single model. The POMDP formulation internalises information constraints and myopic rationality yet explicitly assumes risk neutrality and abstracts from credit and participation costs [5], whereas the behavioural and insurance evidence identifies risk aversion and knowledge constraints as the core of bounded rationality [48]. The two bodies of evidence are complementary but have not been jointly implemented, and the quantitative result on exclusion derives from a configurable simulator rather than field deployment, with the authors themselves describing their estimate as a lower bound [5]. Evidence on data scarcity offers only an analogy: in plant phenotyping, reducing training samples from 10% to 5% degrades prediction error substantially [50], which suggests that model quality deteriorates under data-poor conditions but does not constitute direct evidence about farmer adoption.

A further limitation concerns the explicitness of the paradigm contrast itself. No paper in this body of evidence states the opposition between the rational and the boundedly rational smallholder as a formal proposition; the contrast must be inferred from model assumptions such as myopic best-response and ROI-based payoffs [5] and from the behavioural constructs invoked elsewhere [41]. The closest approximation to an explicit genealogical statement comes from a review distinguishing traditional adoption research, which explains uptake through economic, institutional and household characteristics, from a newer strand incorporating social networks, learning, beliefs, risk and trust [41]. That statement is programmatic rather than tested, and several behavioural claims are borrowed from secondary sources rather than re-estimated in the samples under review.

Taken together, the literature supports six borrowable constructs: myopic best-response with a short horizon as a computable proxy for bounded rationality [5]; risk aversion and limited financial buffers entering the payoff function [48]; private information and unobservable constraints in the form of adverse selection and moral hazard [5]; partial observability combined with multi-season dynamics [5]; social learning, trust and learning costs [57]; and an ROI-oriented rather than yield-oriented objective [61]. Each is individually operationalisable, yet no study integrates risk preference, social learning, partial observability and multi-season dynamics within one adoption mechanism. The open frontier is therefore the construction of a unified risk–social–information–dynamic model, validated in the field rather than only in

simulation, and it is this gap that motivates the next section's examination of how complementary assets and institutional access jointly stratify smallholder response to identical technologies.

8 LAGGING MATURITY OF AGRICULTURAL DATA GOVERNANCE AND STANDARDIZATION

After the gaps revealed by the unified risk–society–information–dynamics model, how complementary assets and institutional access jointly shape smallholders' differentiated responses to the same technology becomes a core question that requires further disaggregation.

Having established how the rational smallholder assumption obscures the heterogeneity of actual farming systems, the review now turns to the institutional and infrastructural conditions under which agricultural data are produced, shared, and reused. This chapter argues that agricultural data governance and standardization remain at a lagging stage of maturity: datasets and benchmarks are highly fragmented, no shared standard has achieved factual authority, and governance itself operates less as an autonomous regime than as a necessary but insufficient dimension embedded within broader AI-ready data frameworks. It further traces how this governance deficit functions as a structural precondition—rather than a parallel concern—for the constrained data accessibility of smallholders. The chapter concludes by showing that these arrangements are matched by systematic weaknesses in how data interventions are evaluated and evidenced, a gap taken up in the following chapter.

8.1 The Absence of Shared Benchmarks and the Fragmentation of Evaluation Criteria

To what extent do agricultural datasets and benchmarks remain fragmented, and does the field possess any shared standard capable of supporting dependable recommendations? The evidence assembled here points to a clear answer: the domain currently lacks a common evaluation substrate, and the datasets that do exist are largely bespoke, author-curated artefacts whose scope and labelling conventions differ too widely to permit cumulative comparison.

The most direct indicator of this condition is the absence of any dataset that functions as a de facto evaluation benchmark. Across the 104 dataset resources identified in the reviewed literature, none is used as an evaluation basis by two or more independent studies; every resource is confined to single-use or niche application. Studies of plant stand counting, nematode cyst phenotyping, active learning for plant phenotyping, and multi-growth-stage weed recognition each construct and evaluate on their own imagery, with no overlap in the underlying data. This pattern is not incidental to individual projects but structural: because no shared corpus anchors reported performance, cross-paper accuracy comparisons are effectively uninterpretable, and the field cannot accumulate evidence in the way benchmark-driven subfields do.

A second dimension of fragmentation concerns scale and provenance. Mainstream agricultural AI datasets are typically one-off products collected and annotated by the authors themselves, and they are small. The corn stand dataset used for deep-learning-based counting, for example, comprises only 211 images captured from three plots over two days (15–16 May 2018), split into 169 training and 42 validation images. Datasets of this size may suffice to demonstrate a proof of concept, yet they cannot support the variance estimates, robustness checks, or external validation that reliable agronomic recommendations require. The contrast with mature vision benchmarks, where tens of thousands of labelled instances are routine, is stark and suggests that the bottleneck is institutional rather than algorithmic.

Label provenance and granularity introduce a further, subtler incompatibility. Annotations are predominantly produced by the authors themselves, and their granularity is typically organ- or function-level rather than the field- or farm-level resolution at which management decisions are made. The mismatch is visible in mapping work, where the absence of publicly available field-boundary data forced the authors to manually digitise 8,176 polygons to serve as ground truth. Such labour-intensive substitution is a symptom of missing shared infrastructure: when no authoritative boundary layer exists, each study manufactures its own reference standard, and comparability across studies is lost at the point of label creation rather than at the point of modelling.

Comparability also degrades over time and across sensors, even when the same instrument and site are retained. A multi-year multispectral study documents severe radiometric incomparability under automatic exposure: the coefficient of determination for predicted total nitrogen uptake falls from above 0.40 under fixed exposure to below 0.13 under automatic exposure, while the blue-band exposure-time distributions differ markedly between 2021 (1.00–1.30 ms at 2x) and 2022 (1.00–2.00 ms at 1x) [62]. This finding is consequential for benchmark design, because it implies that a dataset's utility depends on acquisition protocols that are rarely reported in sufficient detail. Two studies using nominally identical cameras may

therefore be measuring different quantities, and pooling their data without calibration metadata would propagate systematic error rather than reduce it.

Spatial distribution shift compounds the problem in a way that is structural rather than incidental. Deep models require thousands of samples, yet such volumes are frequently unavailable in low- and middle-income countries; moreover, existing studies concentrate on fields of 2–10 hectares, whereas smallholder systems in Bangladesh, Nigeria, and the Indo-Gangetic Plain are dominated by holdings below 0.6 hectares [35]. The consequence is that the datasets underpinning published models systematically under-represent the farms where recommendations would have the greatest welfare impact. A model validated on large mechanised fields cannot be assumed to transfer to fragmented smallholder plots, and the literature offers little evidence either way because the relevant test data are absent.

The gap is not merely one of coverage but of categorical omission. Among 226 identified problem categories, five infrastructure-related classes—temporal mismatch, spatial fragmentation, limited machine readability, unclear governance, and smallholder exclusion—are covered by no agricultural imagery dataset at all, and can only be represented through national administrative data systems [50]. This is a qualitative blind spot rather than a sampling deficiency: the problems most relevant to governance and equity are precisely those for which no benchmark imagery exists, so algorithmic progress on them cannot be measured even in principle.

Even the most advanced national data systems fall short of serving as shared standards. For India, the assessment is explicit that no agricultural dataset currently satisfies all five AI-readiness criteria, with the most developed systems only partially aligned [50]. The Soil Health Card programme illustrates the mechanism of failure: samples are aggregated by grid across multiple farms, eroding persistent plot-level identifiers, and outputs are issued only as static PDFs without JSON, CSV, or API endpoints. Despite 82% farmer awareness, only 48% adoption of recommended fertiliser rates is observed (n=3,600) [50]. The design defect is therefore doubly consequential—it forecloses AI integration and coincides with weak behavioural uptake, although the observational design cannot establish that the former causes the latter.

Taken together, these findings indicate that fragmentation operates at four coupled levels: the absence of any multiply-used benchmark, the small scale and author-specific provenance of individual datasets, the incomparability of labels and radiometry across studies, and the categorical exclusion of infrastructure and smallholder problems from existing corpora. The coupling matters, because each level reinforces the others—without shared benchmarks there is no incentive to standardise labels, and without standardised labels no shared benchmark can be assembled. The result is a field in which reported performance is locally meaningful but globally incommensurable, which in turn weakens the evidential basis for the recommendation systems examined in the following section.

8.2 From Isolated Principle to Operational Constraint: Governance as an Embedded Condition of Smallholder Data Access

Whether data governance in agriculture names a discrete principle or an actionable framework—and how either reading bears on smallholder data accessibility—is the question this subsection addresses. The evidence assembled here supports neither pole in its pure form. Governance emerges as a structured, multi-attribute construct whose operational content is specified with reasonable precision at the level of design principles, yet whose translation into enforceable arrangements for smallholder-facing data systems remains demonstrably incomplete. The argument proceeds from the formal embedding of governance within AI-readiness definitions, through parallel operational frameworks and their reinforcement dynamics, to the maturity gap that separates principle from practice and the accessibility deficits that gap produces.

The most direct evidence against treating governance as an isolated concern comes from the formal definition of AI-ready data, which specifies five co-equal properties: persistent identifiers, machine-accessible formats, temporal alignment, semantic interoperability, and transparent governance [50]. Governance here is not an external ethical overlay but a constitutive constraint that operates alongside the technical attributes; a dataset lacking transparent access rules is, by this definition, not AI-ready regardless of its formatting quality. This framing carries analytical weight because it converts an abstract value into a testable property of data artefacts. The same framework explicitly identifies unclear data governance as a constraint on access and reuse, indicating that the definition was constructed with the access problem in view rather than appended to it. Governance, on this reading, is embedded in the readiness construct itself.

A second, independent framework corroborates the operational character of governance while specifying its content differently. The ethical AI recommendations for digital agriculture organise governance around six principles—fairness, transparency, accountability, sustainability, privacy, and robustness—and translate

them into concrete prescriptions for agricultural technology providers and policymakers, including data portability, interoperability standards, informed consent, and privacy-by-design. The contrast with the AI-readiness account is instructive: where the latter treats governance as one property among five that jointly determine technical usability, the former treats it as a bundle of obligations that jointly determine ethical acceptability. That two frameworks with different purposes converge on governance as a structured, decomposable object—rather than a slogan—strengthens the case that the concept is operationalisable, even as their divergent principle sets show that no single canonical specification has been settled.

The interaction between governance and the other determinants of adoption supplies the mechanism linking framework design to smallholder outcomes. Digital technology adoption among smallholders is characterised as a five-dimensional phenomenon whose components do not operate in isolation but mutually reinforce or weaken one another, with policy and data-institutional support constituting a critical layer; when any dimension is imbalanced, adoption becomes partial, unsustainable, and may even widen the inequality gap [57]. This claim reframes governance from a compliance requirement into a system-level condition: weak data institutions do not merely fail on their own terms but depress the returns to complementary investments in devices, skills, and services. The implication for smallholders is structural rather than incidental—governance deficits propagate through the adoption system rather than remaining confined to a single dimension.

The smallholder-specific consequence of this embedding is stated most starkly in the Indian context, where smallholders constitute 86 percent of farmers and are described as lacking the capacity to compensate for weak data infrastructure, placing them at risk of exclusion [50]. The mechanism deserves emphasis: because governance is one of five jointly binding properties, a smallholder whose data flows through systems with opaque access rules or absent machine-readable consent cannot substitute other resources to achieve usability. Exclusion is thus a derivable consequence of the definitional structure, not an empirical accident. This distinguishes the smallholder problem from a general infrastructure deficit—the constraint binds precisely because governance is non-substitutable within the readiness construct.

Quantitative evidence on insurance access illustrates how governance gaps manifest as measurable uptake failures. Among smallholder farmers in Ghana, digital insurance access and acceptance rates stand at only 14 percent even though 90 percent regard the product as useful, with insufficient knowledge (64 percent) and product unavailability (23 percent) as the dominant barriers [48]. These figures are structurally homologous to the access-rule opacity and missing machine-readable consent diagnosed in the readiness framework [50]: in both cases the failure lies not in perceived value but in the institutional and informational conditions that would convert value into access. The convergence of a demand-side survey and a supply-side data framework on the same bottleneck is among the stronger pieces of evidence in this literature, although the geographic and methodological distance between the two sources means the correspondence should be read as suggestive rather than established.

Maturity evidence tempers any inference that operational frameworks are already functioning as intended. India's Digital Agriculture Mission allocated 2,817 crore to agricultural data infrastructure, yet at the time of writing only 19 states had connected to AgriStack through memoranda of understanding, national deployment had been deferred to 2026–27, and the data ecosystem was assessed as only partially aligned with the requirements of production-grade AI [50]. This is a transitional configuration: top-level design is funded and articulated, but the institutional plumbing that would make governance operational at scale is incomplete. The gap between allocation and connection is itself a governance phenomenon—federal coordination, state incentives, and interoperability agreements are precisely the mechanisms the ethical framework prescribes [63]—which suggests that the binding constraint lies in implementation rather than in conceptual specification.

Fragmentation at the policy level reinforces this reading from a different jurisdiction. A systematic review of digital technology utilisation among South African smallholders reports a lack of reports and policy documents and raises questions about how smallholder digital technology policy is actually developed [15]. Where the Indian case shows a funded programme in transition, the South African case shows an earlier stage in which the policy object itself is not yet documented. Placed side by side, the two cases trace a maturity gradient along which the same governance functions—documentation, coordination, standards—are sequentially absent, and they indicate that the operational frameworks available in the literature have not yet been institutionalised in smallholder-facing settings.

Taken together, the evidence supports a qualified answer to the question posed at the outset. Data governance is not an isolated concept: it is embedded as a co-equal property in formal readiness definitions [50], elaborated as an actionable principle set for providers and policymakers [63], and positioned as a critical layer in adoption systems whose imbalance widens inequality [57]. Its link to smallholder data accessibility is therefore structural, operating through non-substitutable constraints that manifest as mea-

surable access deficits [48]. What remains unresolved is the implementation layer: funded but incomplete deployment [50] and undocumented policy environments [15] show that operationalisation on paper has not yet become operationalisation in practice.

This distinction between conceptual operationalisation and institutional operationalisation carries forward directly into the question of standardisation. If governance principles are specified but not enforced, the standards that would give them effect—for consent, portability, and interoperability—remain voluntary and unevenly adopted, which is the condition under which the labelling and benchmarking fragmentation described above persists. The next subsection takes up this standardisation deficit as the institutional counterpart to the conceptual maturity documented here.

9 SYSTEMATIC GAPS IN EVALUATION SYSTEMS AND EVIDENCE STRUCTURES

If conceptual maturity has not yet automatically translated into operable standards, then the standardization deficit itself constitutes an institutional touchstone for testing whether this conceptual framework can be implemented.

Building on the preceding account of immature data governance and standardization, this chapter turns to the evaluation infrastructures through which agricultural AI claims are warranted, and identifies a structural tension between what the field measures and what its own adoption, energy-efficiency, and field-robustness concerns require. Accuracy-family metrics dominate evaluation practice so thoroughly that these three dimensions occupy almost no comparable metric position, while the evidence base remains at a pre-paradigmatic stage in which supportive and contradictory findings have yet to form a debatable consensus layer. The chapter first characterizes this metric skew, then examines the resulting evidence structures, and finally assesses the constituent elements already available for cross-layer indicators and a corresponding research agenda, whose coverage in trustworthy engineering remains markedly thin.

9.1 Metric Skew and the Comparability of Reported Conclusions

How does the concentration of evaluation metrics on accuracy-oriented measures, alongside the scarcity of energy, latency, and adoption indicators, constrain the comparability of conclusions drawn across studies? The evidence assembled here indicates that this skew is not a neutral reporting habit but an epistemic condition that shapes which claims can be compared, which cannot, and which are silently rendered incommensurable.

Accuracy-family metrics constitute the only evaluation dimension on which the field has converged into a community-wide consensus. Detection and segmentation studies routinely report mAP, AP, PQ, IoU, and F1, and each of these measures is invoked across a substantial cluster of methods rather than in isolation: MatchPlant reports mAP of 40.4% and AP of 85.9% for UAV-based single-plant detection [38], while panoptic segmentation work reports PQ values such as 81.89 [64], and stomatal phenotyping systems adopt the same accuracy vocabulary. This convergence is genuinely useful: it permits direct numerical comparison within the detection and segmentation task family, and it explains why that family exhibits the highest degree of cross-study commensurability in the corpus.

However, convergence on accuracy is not equivalent to adequacy of accuracy. A high-coverage consensus metric can still be a path-dependent second-best standard, and the corpus contains explicit statements to this effect. Genomic prediction work argues that accuracy alone is insufficient to assess generalisation and that a realised selection differential is required to capture decision-relevant performance. Segmentation metrics carry definitional defects of their own: aggregated polygons that fail the IoU ≥ 0.5 criterion are not counted as valid predictions, which may systematically underestimate performance, and correct identification of detections still depends on ground-truth labelling [35]. Radiometric work further shows that R^2 values fall below 0.5 under incomplete canopy closure [62]. The consensus metric, in other words, is shared precisely because it is convenient to compute, not because it has been validated as a sufficient summary of system quality.

In contrast to the dense accuracy cluster, energy and latency indicators are almost entirely absent as shared measures. Latency per inference, energy consumption per inference, frame rate, training time, and reconstruction efficiency each appear in the corpus but remain confined to single studies, well below any threshold that would support cross-study aggregation. Hardware-aware YOLO compression for STM32U5 microcontroller deployment reports efficiency figures that no other study replicates in comparable form [39], and object-centric 3D Gaussian splatting for strawberry reconstruction reports reconstruction efficiency in a protocol specific to that pipeline [17]. The consequence is structural rather than incidental: a reader cannot determine whether a method reporting higher accuracy at unknown cost is preferable to a leaner method, because the cost axis has no common scale.

The adoption dimension presents a different failure mode. Here metrics nominally exist in plurality, but their definitions are mutually incompatible, so that aggregation would be misleading rather than merely incomplete. A survey of smallholder insurance acceptability reports adoption rates around 14% alongside perceived usefulness near 90% [48]; tractor service studies report a 79.7% adoption rate under a different construct [56]; and principal-agent modelling of carbon farming contracts separates large-farm simulated uptake of 87.7% from smallholder uptake of 8.2% [5]. These figures cannot be merged into a single adoption indicator, because they measure different populations, different technologies, and different behavioural constructs. The skew is therefore twofold: adoption is under-measured in technology papers and inconsistently measured where it does appear.

Comparability diverges sharply across task families, and this divergence is best understood through set overlap rather than through reported values. Within object detection, the metric sets used by MatchPlant, boundary-loss panoptic segmentation, and StomaD2 overlap substantially, with Jaccard similarity above 0.5, indicating genuine convergence [38, 64]. For LLM-based decision support, by contrast, evaluation is built from policy alignment ratio, area under the response curve, latency, and human workload [4], from domain-specific conversational evaluation suites, and from expert scoring, with almost no intersection among these sets. Causal adoption studies likewise rely on average treatment effects and related estimands [49], which share essentially no vocabulary with either of the preceding families. The Jaccard similarity across these families approaches zero, which means that the apparent plurality of metrics conceals an absence of shared measurement.

A natural question is whether the missing dimensions are genuinely unmeasurable or merely unmeasured. Several recent studies demonstrate the former is not the case. Pezego-HITL formalises evaluation as a four-objective optimisation over policy alignment, response quality, latency, and human workload, showing that efficiency and human effort can be placed on the same evaluation footing as quality [4]. Principal-agent work introduces a resource-adjusted separation that exposes how cost structures and profit maximisation generate exclusion, making adoption equity a computable quantity rather than a qualitative caveat [5]. PhenoIntel reports uncertainty calibration with expected calibration error of 0. and empirical coverage of 93.6%, demonstrating that reliability can be quantified alongside accuracy [1]. These examples establish that the skew is a choice of the research community rather than a limitation of the phenomena.

Field robustness occupies an intermediate position: it is widely acknowledged but rarely operationalised as a metric. Instead, it surfaces as a limitation statement appended to accuracy results. Studies note species-specific training [32], restriction to a single crop and region [49], and validation confined to a single early season [38]. A review of AI deployment in agriculture reports substantial performance losses when models are transferred to new agro-ecological zones [50]. Because robustness never acquires an independent metric, it cannot enter comparative tables, and a method validated in one season is presented in the same numerical format as one validated across seasons.

The cumulative effect on conclusion comparability is best characterised as a systematic asymmetry in what reported numbers license. Accuracy-dominated reporting supports the inference that method A outperforms method B on a benchmark, but it does not support the inference that A is a better deployment choice, because the cost and adoption axes on which deployment decisions rest are either single-study or definitionally fragmented. Where adoption is measured, incompatible constructs prevent pooling [48, 56]; where efficiency is measured, single-study protocols prevent replication [39]; and where robustness matters most, it is reported as prose rather than as a number [50]. Comparability, then, is high exactly where deployment stakes are lowest.

Two boundary conditions qualify this account. First, the skew is not uniform across the corpus: detection and segmentation form a genuine island of commensurability, and the critique applies with full force only to decision-support and adoption-oriented work. Second, the incompatibility of adoption metrics partly reflects legitimate contextual variation, since smallholder and large-farm settings differ in ways that a single indicator would obscure; the problem is not that contexts differ but that no framework records the contextual parameters alongside the number. Recognising both conditions sharpens rather than weakens the central claim: the field lacks a shared measurement architecture, not merely a shared metric list.

Taken together, the evidence indicates that metric skew degrades comparability through three coupled mechanisms: accuracy consensus creates an illusion of commensurability that extends only within task families; efficiency and latency remain single-study artefacts that cannot be aggregated; and adoption metrics fragment into incompatible constructs that resist pooling. The studies that have built multi-dimensional evaluations demonstrate that these gaps are closable [4, 1], which shifts the question from feasibility to incentive. This raises the further issue of how evidence is structured when metrics are absent altogether, and it is to the architecture of reported evidence—rather than to the metrics themselves—that the following section turns.

9.2 The Architecture of Reported Evidence: Absent Contradiction, Unresolved Tension, and the Missing Temporal Axis

If the preceding discussion established that incentive-compatible design remains under-evidenced, a more fundamental question concerns the structure of the evidence itself: what does it mean for cumulative knowledge when the reported literature contains virtually no recorded disagreement and no temporal ordering at all? Across the six studies examined here, the structured evidence record contains a single supporting relation linking to [52] and no contradicting relation whatsoever. The field therefore presents what is best described as a state of parallel declaration rather than critical dialogue: claims accumulate side by side without being placed in explicit opposition, and the absence of recorded conflict is not evidence of consensus but evidence that disagreement is not represented in our evidence set formally registered.

The single supporting relation is itself instructive, because it is a framing-level endorsement rather than a substantive replication. [65] explicitly positions its work as sharing the same agroecological framing as while conceding that it does not directly extend that study's specific ecological nutrient management method. No cross-reproduction at the level of data, site, or outcome measure accompanies the link. This suggests that even the one positive relation in the record attests to conceptual affinity rather than empirical convergence, and that the density of genuine corroboration in this literature may be considerably lower than a simple count of supporting links would imply.

A sharper contrast emerges when the causal-inference strand is read against the extension literature. [49] report that digitally delivered sowing recommendations raise yields by 12–17%, with an average treatment effect of 372–546 kg/ha against a mean of 3145 kg/ha, and they subject this estimate to four separate refutation tests (placebo, random common cause, subset, and unobserved-confounder sensitivity). [54] reproduce a consistent interval across 171 cotton fields, lending the positive effect external credibility. Read together, these two studies constitute the strongest quantitative agreement in the corpus, yet that agreement was established through independent methodological rigor rather than through any recorded relation between them.

Against this stands an unregistered quantitative conflict. [61] documents that digital agronomic advice raises the likelihood of fertilizer use by a factor of two to five, yet observes no impact on self-reported cotton yields or on satellite-based yield estimates. This null yield result sits in direct tension with the positive average treatment effects above, and the tension is substantive rather than incidental: input intensification without yield response implies either diminishing returns at the margin or a mismatch between the recommended practice and local agronomic conditions. That this conflict appears nowhere in the structured record indicates that the evidence architecture is insensitive to precisely the kind of disagreement most informative for cumulative knowledge.

Internal tensions within individual studies are similarly invisible to the record. [52] concedes that short-term yield outcomes under ecological nutrient management are variable, while [65] acknowledges that the claim of meager yields in agroecological systems is contested and that better performance is not guaranteed. Each of these statements is a hedged admission of uncertainty, and each would ordinarily warrant a recorded qualification or counter-claim; instead, both pass through the evidence structure without generating any opposing relation. The mechanism appears to be that hedged language is read as neutral description rather than as a positioned claim, so qualifications dissolve rather than accumulate.

The scale narrative supplies a further contradiction that the record does not capture. [50] characterizes AI diffusion in Indian agriculture as fragmented and largely pilot-driven, well below the 25–40% adoption rates observed in other sectors, and notes that even the country's most advanced initiatives remain only partially aligned with operational needs. This directly challenges the implicit narrative that digital tools are already effective and scalable, a narrative that the positive causal estimates above could otherwise be taken to support. The two strands are not irreconcilable—yield effects at the plot level and diffusion at the system level are different objects—but the record offers no apparatus for stating that distinction, and so the contradiction remains latent.

The temporal dimension is where the structural gap is most consequential. The only temporal metadata in the corpus comes from the systematic review by [15], which traces a growth from one study in 1994 to twenty-two in 2022 and documents a transition in research focus over that period. Yet every study in the structured record carries an unknown temporal position, which means that the question of whether evaluation metrics have converged, drifted, or fragmented over three decades cannot be posed at all. A field that cannot order its own claims in time cannot distinguish genuine methodological convergence from the successive replacement of incompatible conventions.

This temporal blindness compounds with the metric imbalance documented earlier. The evaluation vocabulary is heavily weighted toward precision and quality, while efficiency-oriented measures appear

only sporadically and are rarely used, as in the scattered reporting of inference latency and reconstruction efficiency by [1], [66] and. Without temporal ordering, one cannot determine whether this imbalance is a stable feature of the field or a transitional artifact of early-stage tooling; the two possibilities carry very different implications for how the gap should be addressed.

What would it take to remedy these deficiencies? The causal-inference studies point toward one answer: disagreement becomes tractable when claims are stated as estimands with explicit identification assumptions, since two estimates of the same quantity can then be compared directly. [49] and [54] demonstrate that this discipline produces comparable quantities even without any coordinating framework. The lesson is that the missing element is not more studies but commensurable reporting: shared outcome definitions, stated estimands, and explicit temporal anchors would allow conflicts such as the fertilizer-versus-yield divergence to be adjudicated rather than merely noted.

A complementary route runs through the review literature. [15] shows that temporal aggregation is feasible when a review deliberately codes study year, focus, and context; the same coding applied at the level of individual claims would supply the temporal axis that is currently absent. Similarly, the pilot-versus-scale contrast drawn by [50] becomes a testable proposition once adoption stage is recorded as a study attribute rather than left implicit. In both cases the remedy is architectural rather than empirical: the evidence already exists, but its relational structure does not.

The boundary conditions of this argument deserve statement. The absence of recorded contradiction does not by itself prove that the underlying literature is uncritical; it may instead reflect the limitations of how the present corpus was assembled, and the six studies examined here are too few to support strong claims about the field as a whole. Hedging is therefore appropriate: the pattern is consistent with a parallel-declaration regime, but it may also be an artifact of coverage. What can be asserted with more confidence is that the unregistered conflicts identified above—between positive yield effects and null yield effects, and between effectiveness claims and diffusion evidence—are real features of the cited studies and would remain invisible under any architecture that records only support.

Taken together, these observations indicate that the central deficiency is not a shortage of findings but an inability to relate them: support without contestation, claims without dates, and metrics without commensurability. The consequence for knowledge accumulation is direct, since a literature that cannot record disagreement cannot correct itself, and one that cannot order its claims in time cannot show whether it is converging. This reframes the gap as one of evidentiary architecture rather than evidentiary quantity, and it motivates the shift, in what follows, from how evidence is structured to how the evaluation targets themselves are chosen and bounded.

9.3 Cross-Layer Indicators and a Research Agenda for Trustworthy Agricultural AI

What indicators should span the layers of agricultural AI systems, and which agenda items follow from them, is the question this section addresses. The preceding analysis established that the central deficit is architectural rather than quantitative: the field does not lack measurements but lacks measurements that connect assumptions, problems, and limitations across layers. A cross-layer indicator regime therefore has to satisfy three conditions simultaneously—it must trace trust deficits along the chain from stated assumption to observed problem to acknowledged limitation, it must be comparable across datasets and reporting conventions, and it must be cheap enough to be adopted by the small teams that dominate agricultural AI development. The evidence assembled here suggests that no current proposal meets all three, but that several partial solutions already exist and can be composed into a coherent agenda.

The first agenda item concerns assumption auditing. Cerealia makes explicit that agricultural decision systems routinely presuppose that sensor readings are neither faulty nor under attack, and that this presupposition fails in practice, which locates the origin of security and robustness gaps not in component failures but in unstated modelling assumptions. This is a different diagnosis from the one implied by benchmark-centric evaluation: a system can score well on a held-out test set while resting on an assumption that the deployment context violates. A cross-layer indicator should therefore record, for each system, the assumptions under which its reported performance holds, and should flag any downstream limitation that can be traced to an assumption left unexamined. The proposal is modest but consequential, because it converts an implicit design choice into an auditable object.

The second item concerns adoption, where the evidence is unusually strong and unusually narrow. Pezego-HITL reports an AI trust scale with a Cronbach alpha of 0.944 and an acceptance rate of 85.9% for AI recommendations, with only 14.1% requiring substantive correction [4]. These are precisely the kind of human-facing indicators that technology-centred evaluations omit, and their psychometric grounding distinguishes them from ad hoc satisfaction surveys. However, a single deployment in a single national

extension context cannot establish that the instrument transfers; the alpha value demonstrates internal consistency, not external validity. The agenda item is thus to treat trust and acceptance as measured constructs with reported reliability, while requiring replication across contexts before they are used comparatively.

Field-level value admits a stronger form of quantification, and the contrast with acceptance metrics is instructive. Causal analyses of digital agriculture recommendations find that fields following sowing advice yield 12–17% higher yields (372–546 kg/ha against a mean of 3145 kg/ha), and the estimates survive four refutation tests [49, 54]. The methodological difference matters: acceptance measures what farmers say or do, whereas causal estimates measure what changes when advice is followed, and only the latter supports claims about welfare effects. Pairing the two yields a cross-layer indicator in which stated trust, observed uptake, and realised outcome are reported together, allowing discrepancies—high acceptance without measurable yield gain, or the reverse—to become visible rather than averaged away.

Robustness indicators expose a comparability failure that is more acute than the absence of metrics. MatchPlant reports an average precision of 85.9% at IoU 0.5 for single-plant detection, while a convolutional stand-counting approach reports MAE 1.73 and RMSE 2.46 [38, 33]. Both are legitimate field-robustness measurements, yet they are computed on different datasets with different annotation conventions and different target quantities, so neither can adjudicate the other. The problem is not that the numbers are weak but that they are incommensurable, and a cross-layer regime must therefore standardise the reporting tuple—dataset provenance, annotation protocol, and error definition—before any leaderboard comparison is meaningful. Absent such standardisation, aggregating reported robustness across studies would manufacture precision that the underlying evidence does not support.

Auditability, by contrast, has been engineered into a measurable primitive, which shows that governance properties need not remain qualitative. Pezego-HITL employs a critique agent that checks recommendations against Ghana EPA registration, dosage thresholds, and regional prohibitions, achieving a policy compliance rate of 0.94 [4]. This is a notable result because compliance is verified against an external registry rather than against model self-report, which makes the indicator resistant to the failure mode it is meant to detect. The same design pattern—an external, checkable ground truth consulted at inference time—generalises to other governance properties, and its inclusion in a cross-layer regime would let compliance be reported alongside accuracy rather than in a separate ethical appendix.

Factual consistency for language-model advisors requires its own instrument, and the available evidence indicates that current performance is far from adequate. DG-EVAL applies atomic fact verification against expert ground truth and reports an F1 of only 47.1% for the vegetable category [46]. This figure is best read as a calibration of the problem rather than as a verdict on any single system, since the category-level breakdown suggests substantial variation across domains. Related work on scaling advisory services to smallholders reaches compatible conclusions about the difficulty of maintaining factual reliability under broad coverage [44]. A cross-layer indicator should therefore report hallucination and factual consistency per domain rather than in aggregate, because an aggregate score can conceal exactly the category-level weakness that matters most to a farmer asking about a specific crop.

Governance maturity presents the sharpest asymmetry in the assembled evidence. Privacy and data-ownership primitives are addressed explicitly by only one study in our evidence set, and AI fairness and bias by the same single study, which itself concedes that algorithmic intervention cannot resolve fairness problems. Set against the comparatively mature robustness and auditability indicators discussed above, this indicates that the governance layer of any cross-layer regime is currently emerging rather than established. The honest agenda item is therefore not to propose governance metrics as if they were settled, but to mark them as underdeveloped and to specify what evidence would be needed to mature them—multi-site studies, disaggregated outcome reporting, and explicit treatment of data ownership in deployment agreements.

A further item concerns the temporal ordering of claims, which the preceding discussion of evidentiary architecture identified as decisive. An evaluation regime that records only current performance cannot show whether a system is converging, and one that records only aggregate outcomes cannot show whether gains persist after the novelty of a deployment fades. The causal studies cited above partially address this by estimating effects over a defined period and subjecting them to refutation tests [54], but the trust and acceptance instruments are typically administered once. A cross-layer indicator should therefore specify a reporting cadence, so that trust, uptake, and outcome are measured at matched intervals and trends can be distinguished from levels.

These items compose into a research agenda with four entries. The first is assumption disclosure: every reported result should be accompanied by the conditions under which it holds, following the diagnosis that unstated assumptions are the root of robustness and security gaps [42]. The second is paired human

and outcome measurement, joining psychometrically validated trust instruments with causal estimates of field-level effect [4, 49]. The third is reporting standardisation for robustness, without which cross-study comparison remains illusory [38, 33]. The fourth is domain-disaggregated factual consistency, since aggregate scores obscure the category-level failures that most affect users [46].

Two boundaries on this agenda deserve emphasis. First, the governance entries rest on a single source and should be presented as open problems rather than as recommendations with an evidence base. Second, the quantitative anchors cited throughout derive from specific crops, regions, and deployment arrangements, and their transferability is untested; the trust scale's high internal consistency, for instance, says nothing about whether the same items measure the same construct elsewhere [4]. A cross-layer regime that ignored these boundaries would repeat, at the level of meta-evaluation, the very overgeneralisation it is designed to catch.

Taken together, the evidence supports a cross-layer indicator regime organised around assumption disclosure, paired trust-and-outcome measurement, standardised robustness reporting, externally verified compliance, and domain-disaggregated factual consistency, with governance metrics explicitly flagged as immature. Such a regime would not by itself close the trust gap, but it would make the gap measurable at the layer where it originates rather than only where it surfaces. This raises the further question of how such indicators can be embedded in the incentive structures of research funding and publication, which is where the discussion turns next.

10 DISCUSSION

The five research questions examined in this review were posed independently, yet their answers converge on a small number of structural findings that no single question could have surfaced alone. Two cross-cutting threads run through the corpus. The first is an architecture–task mismatch: the technical supply side optimizes for benchmark accuracy on curated data, while the demand side requires robust, low-cost, language-accessible inference under field conditions that the benchmarks do not represent. The second is an evaluation and evidence fragmentation that makes the field's cumulative claims difficult to assemble: metrics, datasets, and adoption constructs are each locally reasonable but globally incommensurable. The remaining paragraphs trace how the evidence for one question reinforces, constrains, or contradicts the conclusions of another.

The supply-side findings of RQ1 and the evaluation-structure findings of RQ5 are mutually reinforcing in a way that reframes both. RQ1 reports an asymmetric readiness picture: language accessibility is documented and partly quantified, with provider coverage uneven across Hindi, Telugu, and Odia and no single model strong on all three [2], while on-device inference remains an acknowledged open problem whose performance and latency trade-offs are described as unclear [55]. RQ5 explains why this asymmetry persists: energy and latency metrics are used by only a single paper in the entire metric inventory, and accuracy metrics are the only dimension on which the community converges [39]. The absence of on-device feasibility measurements is therefore not an oversight of individual studies but a predictable consequence of an evaluation regime that has no routine place for energy, latency, or field robustness. Conversely, the RQ1 finding that language coverage is comparatively well measured is consistent with RQ5's observation that accuracy-adjacent metrics dominate—word error rate is legible to the prevailing measurement culture in a way that joules per inference is not.

RQ4 and RQ5 jointly constrain how much weight the field's comparative performance claims can bear. RQ4 finds that none of the 104 audited dataset nodes is used as an evaluation benchmark by two or more papers, yielding zero de facto standards and zero commonly shared datasets [33, 67]. RQ5 finds that even where metrics converge within a task family, cross-family comparability collapses: object detection metrics show substantial overlap, whereas LLM decision-support and causal adoption evaluations share almost no metric vocabulary [4, 46]. Read together, these results imply that reported performance differences—whether in mean average precision or in intersection-over-union—are comparisons across different datasets, imaging conditions, and labeling protocols rather than measurements of method quality [67, 35]. This compounds the RQ1 readiness gap: a practitioner seeking a field-deployable advisor cannot adjudicate between systems on published evidence, because neither the data nor the metrics provide a common yardstick. The RQ4 finding that datasets are one-off research artifacts, often tiny and author-labeled [33], further undercuts any inference from benchmark accuracy to farm-level reliability.

The adoption-side questions, RQ2 and RQ3, expose a tension between methodological rigor and the realities of smallholder heterogeneity that qualifies the confidence of the causal literature. RQ2 finds that only two of six adoption studies perform causal inference, and that these two are successive iterations by the same team on the same cotton-planting dataset, reporting a yield gain of 12–17 percent under a back-door identification strategy validated by placebo, random-confounder, and subset robustness checks [49, 54].

RQ3 shows why the identification assumptions are fragile in this setting: financial constraints bind across all farm categories as the top-ranked constraint [56], credit access separates adopters from non-adopters at roughly 50 versus 23 percent [11], and a profit-maximizing aggregator achieves 87.7 percent realized adoption among large farms against 8.2 percent among smallholders [5]. The causal estimates of RQ2 are therefore internally credible but externally narrow: to sustain identification, the analysis treats cooperative management as constant and omits agricultural-knowledge nodes, conceding limited external validity [54]. RQ3 supplies the substantive reason this matters—the constraints that drive non-adoption are precisely the ones the frictionless-optimizer assumption removes from the model, so structural non-adoption risks being misread as preference. The insurance evidence sharpens the point: 90 percent of farmers find insurance useful but only 14 percent have purchased it, with lack of knowledge and product unavailability dominating cost as reasons [48]. Adoption failure here is an access and information phenomenon, not a demand phenomenon, which is exactly what a model without credit, tenure, and participation costs cannot represent.

Across questions, three patterns deserve emphasis. There is broad consensus that the binding constraints on smallholder AI adoption are institutional and infrastructural rather than agronomic: language coverage [44], credit and insurance access [11, 48], and data standardization [33] recur as first-order barriers. There is a genuine contradiction between the modeling assumptions of mainstream adoption and contract models and the empirical constraint structure documented in RQ3, a contradiction that the causal literature of RQ2 acknowledges but does not resolve. And there is tight coupling between the measurement regime diagnosed in RQ5 and every other question: because energy, latency, field robustness, and adoption are not routinely measured, the readiness, comparability, and adoption claims of RQ1, RQ2, and RQ4 cannot be independently verified. The questions are thus not five separate literatures but one system in which evaluation choices propagate into every substantive conclusion.

The reusable contribution of this review is threefold. First, it offers a classification framework that separates technical readiness into measured and merely acknowledged dimensions, showing that accessibility constraints are not homogeneous and that the unmeasured dimensions cluster around deployment rather than development. Second, it proposes a coordinate system for evidence comparability that plots studies jointly by dataset standardization and metric convergence, making explicit why cross-paper performance claims are currently uninterpretable and what would be required to make them otherwise. Third, it identifies a design dimension missing from existing adoption models: the treatment of financial, infrastructural, and informational constraints as endogenous structural features rather than as noise around a frictionless optimum. Together these contributions convert the fragmented findings of the individual questions into an agenda in which evaluation design, data governance, and adoption modeling are treated as a single coupled problem rather than as independent research streams.

11 FUTURE RESEARCH

Future Research

Now that the scattered research questions have been consolidated into an agenda in which evaluation design, data governance, and adoption modeling are mutually coupled, the next task is to identify the questions in this agenda that remain unanswered and to outline actionable research directions.

A balanced reading of the current evidence base reveals that each of the five research questions is partially answered but none is settled, and that the gaps themselves define a tractable agenda. On evaluation, the remote-island assistant work reports latency of roughly 15–20 seconds for simple queries and about one minute for multi-tool queries [45], yet these remain anecdotal observations rather than measured outcomes, with answer quality, grounding faithfulness, latency, and user satisfaction all deferred to future work; the P-EVAL protocol is a rare exception that formalises latency, achieving a 59.6% cache reuse rate and reducing P95 latency from 28.6 s to 12.9 s while reporting PAR = 0.94 and AUR = 0.95, although its evidence rests on 1,240 simulated queries. The opportunity is therefore clear: instrumented field evaluation that couples retrieval-grounded generation with causal outcome tracking would convert anecdotal performance claims into defensible benchmarks. Complementary phenotyping pipelines show that quantitative, reproducible measurement is achievable in agricultural settings and can serve as a methodological template [26].

A critical perspective cautions that several strands of evidence are too thin to support strong conclusions. The diagnostic review of AI scaling in Indian agriculture locates the binding constraint in data infrastructure rather than algorithmic capability, noting that 86% of smallholders lack the capacity to compensate for weak data infrastructure and thus face systematic exclusion from data-driven tools [35]; this is a structural claim built largely on secondary synthesis rather than primary measurement. Similarly, simulation evidence on pooled contracts shows that a profit-maximising aggregator achieves 87.7% of at-

tainable adoption among large farms but only 8.2% among smallholders [5], an exclusion gap that exceeds what cost structure alone explains. Because these findings derive from simulation and review rather than deployed systems, they should be read as existence proofs and hypotheses, not as settled magnitudes.

From a synthesised standpoint, a single lever connects the disparate gaps: the endogenous treatment of adoption constraints. The POMDP formulation explicitly models private adoption costs, unobservable effort, and partial observability of the aggregate, and assumes season-by-season myopic best response under short horizons, making it the most computationally concrete realisation of bounded-rational adoption available [5]. Its counterfactual is instructive—allocating monitoring costs by area eliminates the size gradient [5]—demonstrating that treating constraints as exogenous covariates systematically understates exclusion. This same logic extends to evaluation: if latency, cost, and data readiness are treated as external parameters rather than modelled quantities, both system design and policy analysis will misstate who can actually use a tool.

The data foundation remains the least contested but most consequential gap. The scarcity of diverse public benchmark datasets designed for varied agricultural contexts is a field-level consensus, and AI-readiness has been formally defined in terms of structured, standardised, and machine-accessible datasets that can be ingested into AI pipelines [50]. Without such corpora, grounding faithfulness cannot be audited, latency–quality trade-offs cannot be compared across systems, and equity analyses cannot be replicated. Building shared, multilingual, regionally stratified benchmarks is thus a prerequisite for progress on every other question rather than a parallel concern.

Finally, unresolved quantitative conflicts demand reconciliation before strong causal claims are advanced. Digital agronomic advice has been associated with a two- to five-fold increase in the likelihood of fertiliser use, yet the same work observed no impact on self-reported cotton yields or satellite-based yields [45], while ecological nutrient management reports variable short-term yield outcomes [52] and agroecological systems are acknowledged to provide meagre returns in some settings. These tensions suggest that adoption, input intensity, and productivity are distinct outcomes that current designs conflate. A research agenda that pairs policy-grounded language models with causal evaluation, explicit constraint modelling, and shared benchmarks would let the field move from documenting gaps to closing them.

12 LIMITATIONS

The evidence base combines an automated retrieval pool with manual supplementation, and this combination bounds what the review can claim to cover. The automated searches privileged indexed venues and English-language sources, so work published in other languages, in regional outlets, or in non-indexed technical reports is underrepresented; manual supplementation partially offsets this gap but follows the authors' own judgment of relevance and is therefore not exhaustive. The literature search closed at a fixed date, and subsequent publications are not reflected. Coverage should thus be read as representative of the retrieved and supplemented pool rather than as a complete census of the field.

Having outlined an agenda to bridge gaps through causal evaluation, constraint modeling, and shared benchmarks, we must now examine how the evidence base supporting this agenda bounds its conclusions.

Model-assisted extraction and drafting introduced efficiencies that we sought to make verifiable rather than to conceal. Every substantive claim is annotated with the supporting references on which it rests, and the intermediate analytical record—screening decisions, extraction tables, and coding rationale—is retained so that readers can trace how each statement was derived. These measures support auditability but do not eliminate residual error: individual extraction inaccuracies and misattributed details may persist despite repeated checking, and claims should be verified against the cited primary sources before being relied upon.

The review is deliberately scoped to its stated topic. Adjacent areas are included only where they intersect directly with the central questions, and are otherwise excluded even when they bear loosely on the subject. Findings therefore characterize the defined scope and should not be extended to neighboring domains without independent evidence.

13 CONCLUSION

This review set out to explain why AI technology supply remains insufficiently field-ready for smallholders, and the evidence synthesized across the five thematic strands converges on a consistent answer. On accessibility, conversational and LLM-based advisory systems have been extensively documented for low-resource languages, with quantified word error rates, language coverage audits, and measured ASR disparities; yet evaluation of offline, low-compute deployment is markedly asymmetric, with energy and latency routinely self-reported by compression-oriented work while field robustness is almost systematically unmeasured. On

the pathway from perception to advice, multi-agent and neuro-symbolic systems have matured through sensing and recommendation generation but stop short of the adoption stage. On conversion, only two of six studies in the evidence matrix implement causal inference, adoption cost is largely unmodeled by technology-capability-oriented work, and supply-side feasibility claims and adoption-side causal claims are never jointly tested on the same technology and decision point—the bridging failure is one of object misalignment rather than missing evidence on either side.

Only after it has been clarified that the evidence boundary covers solely the adjacent domains that directly intersect with the core question do the answers converged from the various thematic threads possess an efficacy that can be cautiously interpreted; following this delimitation, the evidence from the five threads across dimensions such as accessibility, suitability, and adoption begins to point toward the same conclusion.

On heterogeneity, the matrix supplies an operable inventory of theoretical constructs for explicitly modeling bounded-rational adoption under constraints, indicating that the substantive shift is methodological: from treating constraints as exogenous covariates in expected-utility models toward constraint-bound decision models. On data governance, an audit of 104 dataset nodes and 84 papers reveals extreme fragmentation with no de facto standard, and governance emerges as a necessary but insufficient dimension nested within a broader AI-ready data framework. On evaluation, precision-type metrics dominate absolutely, the evidence base contains zero contradicting and only one supporting relation with time evolution entirely unknown, and this configuration diagnoses a pre-paradigmatic stage of evidence accumulation rather than a data gap.

Taken together, these findings reframe the field's central problem: the bottleneck is not a shortage of capable AI but a structural mismatch between where evidence is produced and where adoption is decided. Researchers should therefore design studies that bind supply-side technical claims and adoption-side causal claims to the same technology and decision point, and treat field robustness and constraint-bound farmer decision models as first-class measured variables rather than discussion-section caveats. Engineers, in turn, should report energy, latency, and field robustness as a single deployment profile and co-design data and evaluation standards with the smallholder contexts in which their systems are expected to operate.

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